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Unit 8: Skeletal System

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Learning Objectives

When you have completed this unit, you should be able to:

- Identify bones and markings of the axial skeleton.
- Identify bones and markings of the appendicular skeleton.
- Build a skeleton using disarticulated bones.

Unit Overview

The **skeletal system** consists of the bones, the joints, and their associated cartilages. Its two divisions are the axial skeleton and the appendicular skeleton (**Figure 8.3**). The **axial skeleton** is composed of the bones of the head, neck, and trunk—specifically, the cranial bones, the facial bones, the vertebral column, the hyoid bone, the sternum, and the ribs (shaded purple in **Figure 8.3**). The **appendicular skeleton** (*append-* = "appendage") consists of the bones shaded "bone-colored" in **Figure 8.3**: those of the upper limbs, the lower limbs, the pectoral girdle (the bones forming the shoulder joint), and part of the pelvic girdle (the bones forming the pelvis and hip joint).

In this unit, we explore the anatomy of the bones and bone markings of the skeletal system, which serves as a foundation for later chapters. For example, the radial and ulnar arteries parallel the radius and the ulna, and the frontal, parietal, temporal, and occipital lobes of the brain are named for the cranial bones under which they are located.

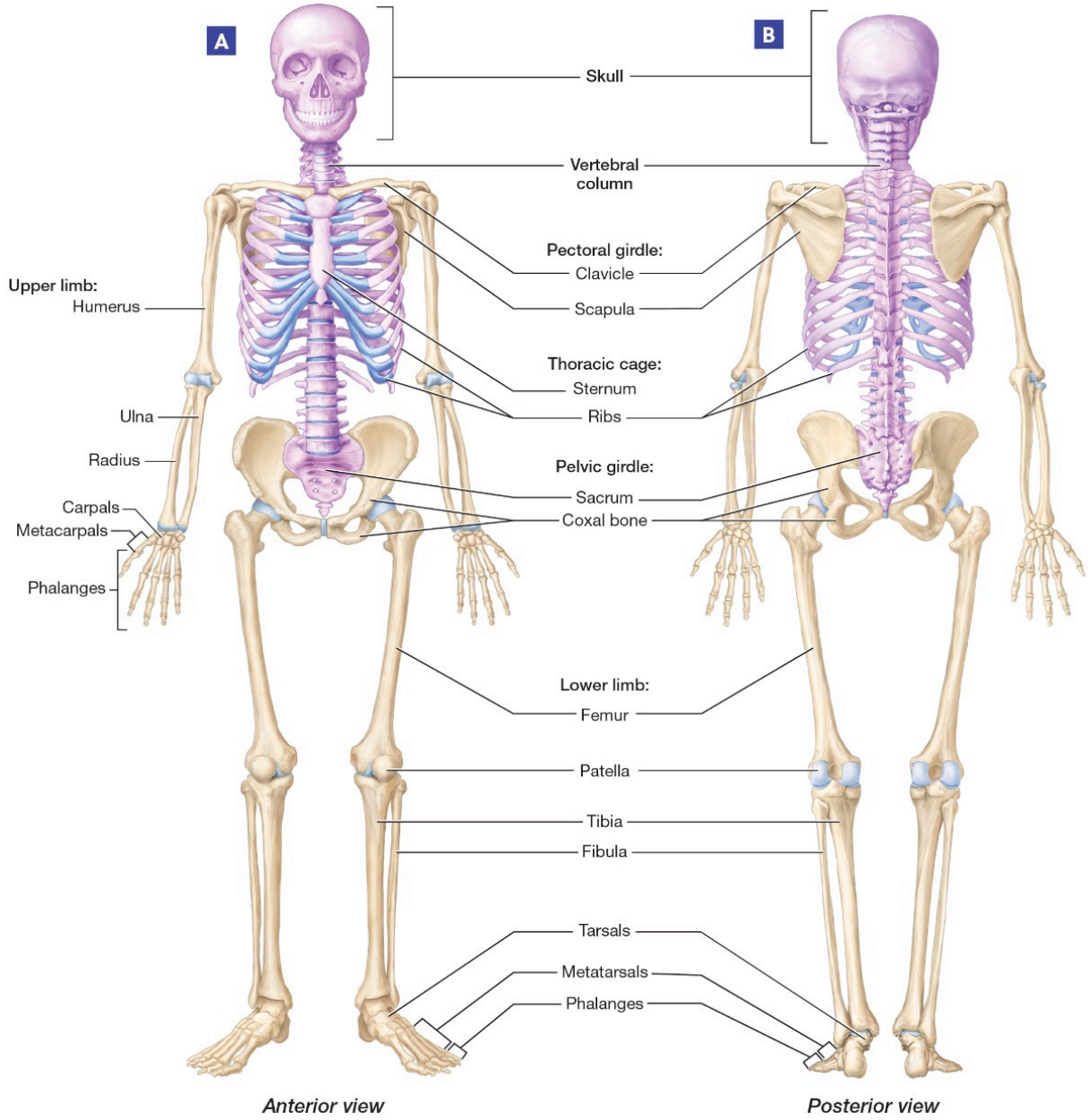


Figure 8.3 The axial and appendicular skeletons: (**A**) anterior view; (**B**) posterior view.

Coloring and Labeling Structures

Bones of the Skull

Color the structures of the skull in **Figure 8.1**, and label them with the terms from Exercise 8-1. Use your text and Exercise 8-1 in this unit for reference.

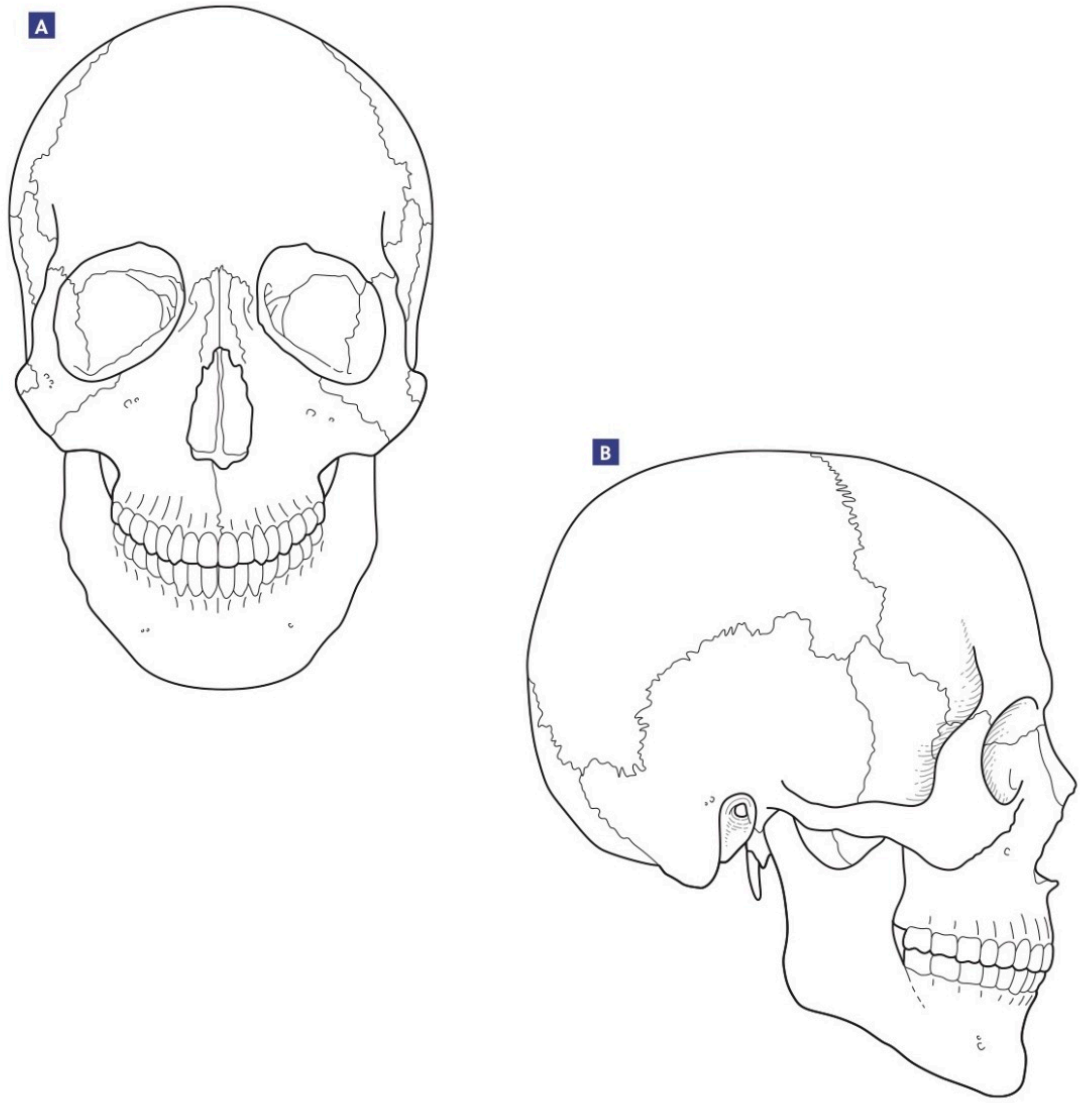


Figure 8.1 (A) A diagram of an anterior view of the skull **(B)** A diagram of a lateral view of the skull; *(continues)*

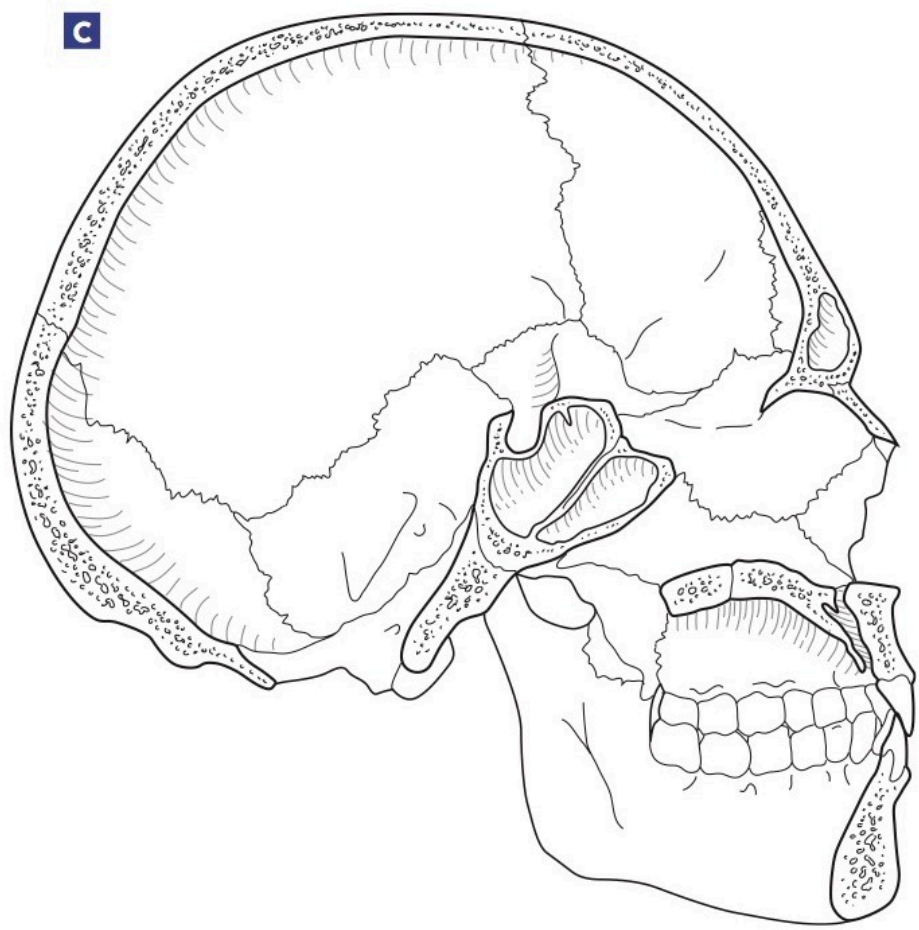
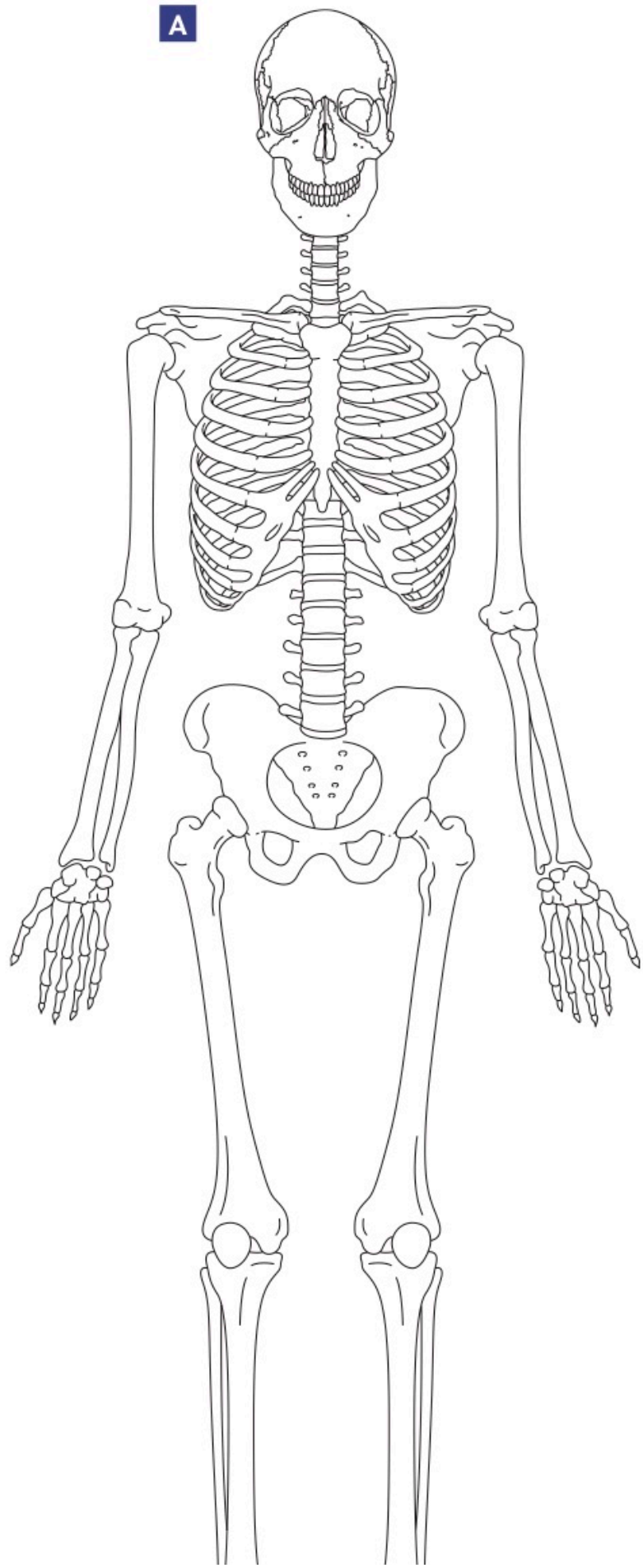


Figure 8.1 Skull **(C)** (cont.) A diagram of a midsagittal section of the skull.

Whole Skeleton

Color the structures of the skull in **Figure 8.2**, and label them with the terms from Exercise 8-1 to 8-3. Use your text and Exercise 8-1 to 8-3 in this unit for reference.


A

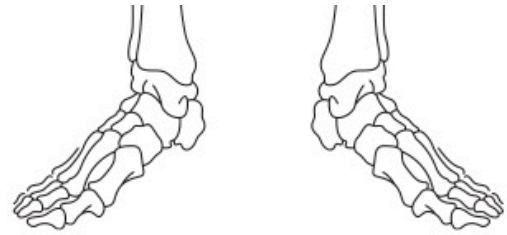
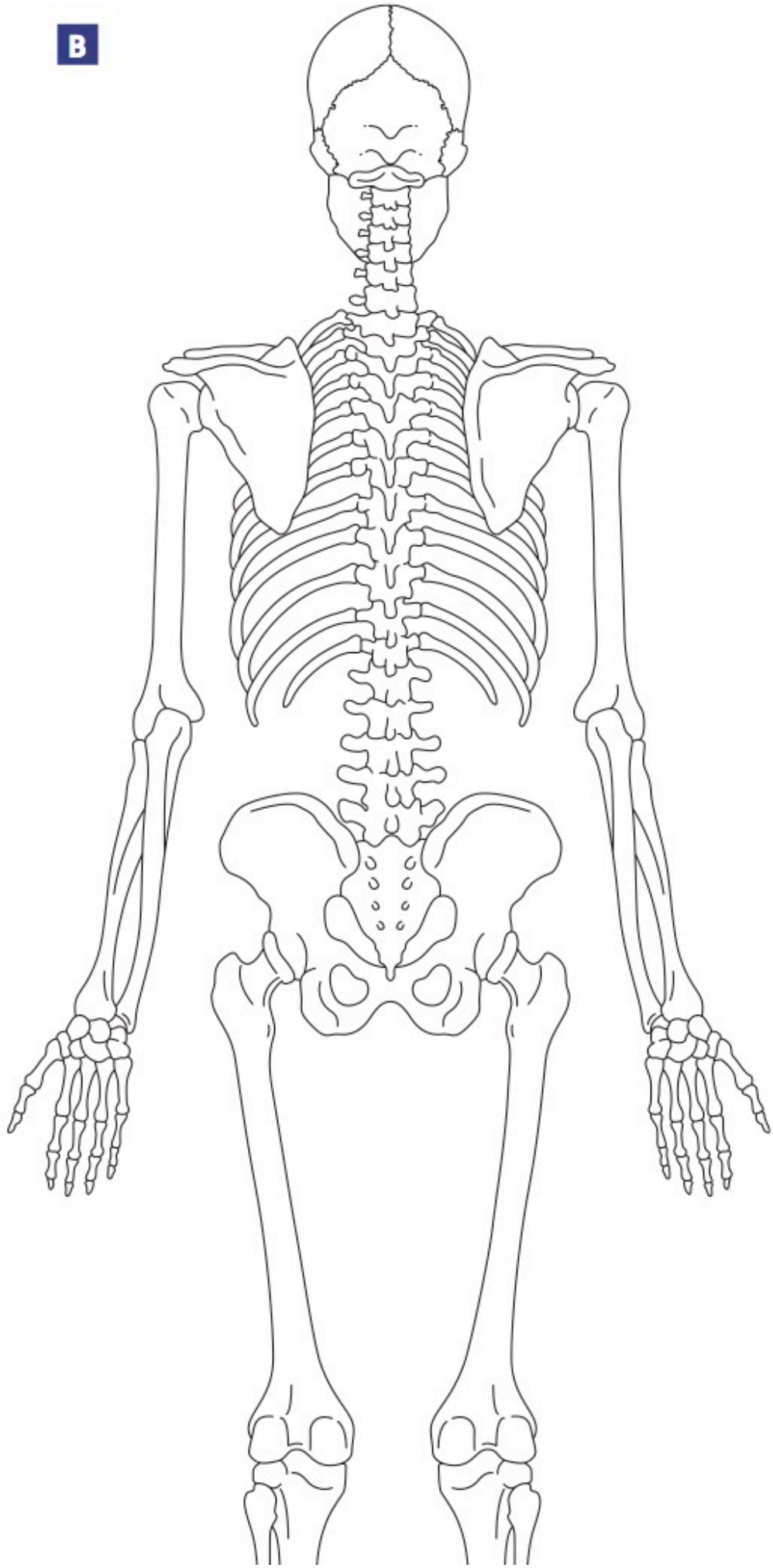


Figure 8.2 Skeleton: **(A)** A diagram of a human skeleton from an anterior view; (*continues*)


B

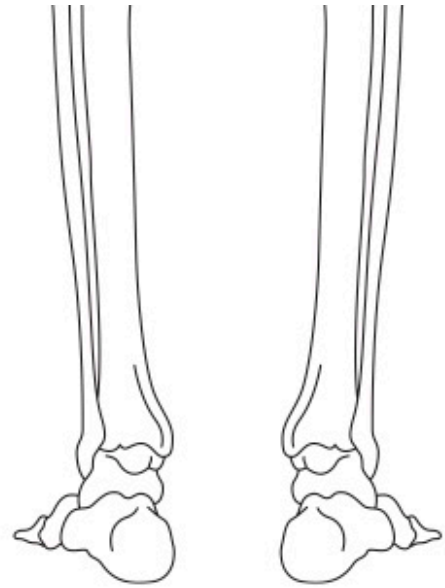


Figure 8.2 Skeleton (cont.) **(B)** A diagram of a human skeleton from a posterior view.

Exercise 8-1 Anatomy of the Skull

Materials

- Adult skulls, whole and sectioned
- Fetal skull

Introduction

The skull is composed of two classes of bones: the cranial bones and the facial bones (**Figures 8.4–8.10**). The eight **cranial bones** (KRAY-nee-uhl; *cran-* = "skull") encase the brain and together form the **calvaria** (kal-VEHR-ee-uh; also known as the *cranial vault*; *calvaria* = "skull"). The calvaria consists of several of the cranial bones joined at immovable joints called **sutures** (SOO-tchurz; *sutur-* = "sew"). These bones also form the **cranial base**, which is made up of the **anterior**, **middle**, and **posterior cranial fossae**—indentations that support the brain (visible in **Figure 8.7**). The cranial bones include the following:

1. Frontal bone;

2. Parietal bones (2);
3. Temporal bones (2);
4. Occipital bone;
5. Sphenoid bone; and
6. Ethmoid bone.

The 14 **facial bones** form the framework for the face, provide openings for ventilation and eating, and form cavities for the sense organs. Several of the facial bones are located deeper in the skull, and you will want to refer to several different figures (noted with each view) to best locate them and appreciate their structures. The facial bones include the:

1. Mandible;
2. Maxillae (2);
3. Lacrimal bones (2);
4. Nasal bones (2);
5. Vomer;
6. Inferior nasal conchae (2);
7. Palatine bones (2); and
8. Zygomatic bones (2).

What follows is a guide to which cranial and facial bones are visible in each view of the skull. At the end is a gallery featuring photographs of different views of the skull (for **Figures 8.14–8.18**).

Anterior View (Figure 8.4)

Cranial Bones

1. The **frontal bone** (*frons* = “forehead”)forms the anterior portion of the cranium, the superior part of the orbit (where the eyeball is housed), and the anterior cranial fossa. Here you can see the smooth area between the eyes called the **glabella** (*glab-* = “smooth”). Over the

anterior ridge of the orbit, an area known as the **supraorbital margin** (*supra-* = "above"), are two small holes called the **supraorbital foramina**.

2. The paired **parietal bones** (puh-RY-ih-tuhl; *pariet-* = "wall") form the superior and part of the lateral walls of the cranium. They are visible on the lateral sides of the anterior view.
3. The paired **temporal bones** form the lateral walls of the cranium. Like the parietal bones, they are visible on the lateral sides of the anterior view.
4. The posterior cranial bone is the **occipital bone** (ahk-SIP-ih-tuhl*; occiput* = "back of the head"). It is not visible in the anterior view.
5. The butterfly-shaped **sphenoid bone** (SFEE-noyd; *sphen-* = "wedge") is posterior to the frontal bone on the interior part of the skull. In **Figure 8.4**, note the **superior orbital fissure**, which is a slit between the greater and lesser wings through which nerves pass.
6. The complex **ethmoid bone** is the deepest cranial bone. From the anterior view, it is partially visible in the medial wall of the orbit.

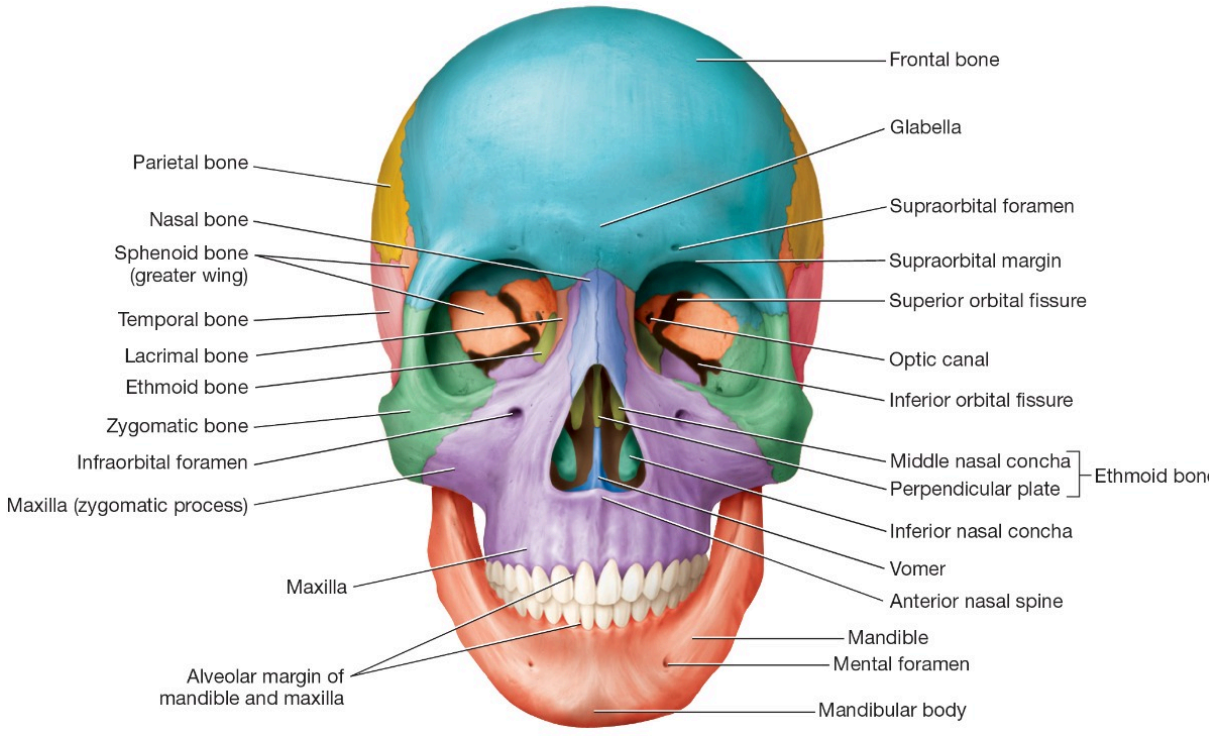


Figure 8.4 Anterior view of the skull.

Facial Bones

1. The **mandible**, or the lower jaw bone, consists of a central **body** and two “arms” called the **mandibular rami** (RAY-mee*; ramus* = “branch”). As you can see in the figure, the lower teeth are attached to the mandibular body along a border called the **alveolar margin** (al-vee-OH-luhr; *alveolus* = “small cavity”). On each side of the lateral mandibular body, we find a small hole called the **mental foramen** (*mentus* = “chin”).
2. The two fused **maxillae** (mak-SILL-ee) are the upper jaw bones. They form the inferior wall of the orbit, part of the lateral wall of the nasal cavity, and part of the superior wall of the oral cavity. Inside the orbit, where the maxilla meets the sphenoid bone, we find a slit known as the **inferior orbital fissure**, which serves as a passageway for nerves and blood vessels.Just inferior to the orbit is a small hole called the **infraorbital foramen** (*infra-* = “below”). Laterally, the maxillae have projections called **zygomatic processes** that form part of the cheekbone. Like the mandible, the maxillae also have an alveolar margin along which the upper teeth are attached.
3. The tiny **lacrimal bones** (LAK-ruh-muhl; *lacrima* = “tear”) are visible in the medial part of the orbit, where they form part of the structure that drains tears produced by the lacrimal gland of the eye.
4. The two **nasal bones** form the anterior framework of the bridge of the nose.
5. The single **vomer** (VOH-muhr) forms the inferior portion of the bony nasal septum. It is visible in the midline of the nasal cavity.
6. The small **inferior nasal conchae** form part of the lateral walls of the nasal cavity.
7. The two **palatine bones** (PAL-uh-ty’n) form the posterior part of the hard palate and the posterolateral walls of the nasal cavity. They are not visible in the anterior view.
8. The two **zygomatic bones** (zy-goh-MAT-ik) form the bulk of the cheek and a significant portion of the “cheekbone” or zygomatic arch.

The orbit and the nasal cavity are complicated structures with contributions from several bones. The **orbit** is formed by parts of seven bones: the frontal bone, maxilla, sphenoid bone, ethmoid bone, lacrimal bone, zygomatic bone, and a tiny piece of the palatine bone. The **nasal cavity** is formed by parts of six bones: the ethmoid bone, maxillae, palatine bones, inferior nasal conchae, sphenoid bone, and vomer (refer to **Figures 8.4 and 8.8**).

Lateral View (Figure 8.5)

Cranial Bones

1. The frontal, ethmoid, and sphenoid bones are visible in this view.
2. Here you can see that the parietal bones articulate with one another and many other cranial bones at sutures: They meet the frontal bone at the **coronal suture** (*corona* = "crown"); they meet the occipital bone at the **lambdoid suture** (LAM-doyd); and they meet the temporal bones at the **squamous sutures**.
3. Notice that each temporal bone has a complex shape with four general regions.

a. The flat **squamous region** is the temporal bone's most lateral portion. As its name implies, it is largely flat; however, it does contain a projection called the **zygomatic process** that forms part of the *zygomatic arch* or cheekbone. Just inferior to the zygomatic process is an indentation, the **mandibular fossa**, which forms the *temporomandibular joint*—or the jaw joint—with the mandible.

b. The temporal bone's inferior region is called the **tympanic region** (tim-PAN-ik; *tympan-* = "drum"). It houses the **external acoustic (auditory) meatus**, an opening to the *external auditory canal*, which leads to the middle and inner ear. Also in this region is a needlelike projection called the **styloid process** (STY-loyd; *stylus* = "pen"). Posterior to the tympanic region is the **mastoid region**, which contains the large **mastoid process**.

Inside the mastoid process are numerous tiny sinuses known as *mastoid air cells*.

c. The temporal bone's fourth region, the *petrous region*, is seen in **Figure 8.6** and is discussed with that view.

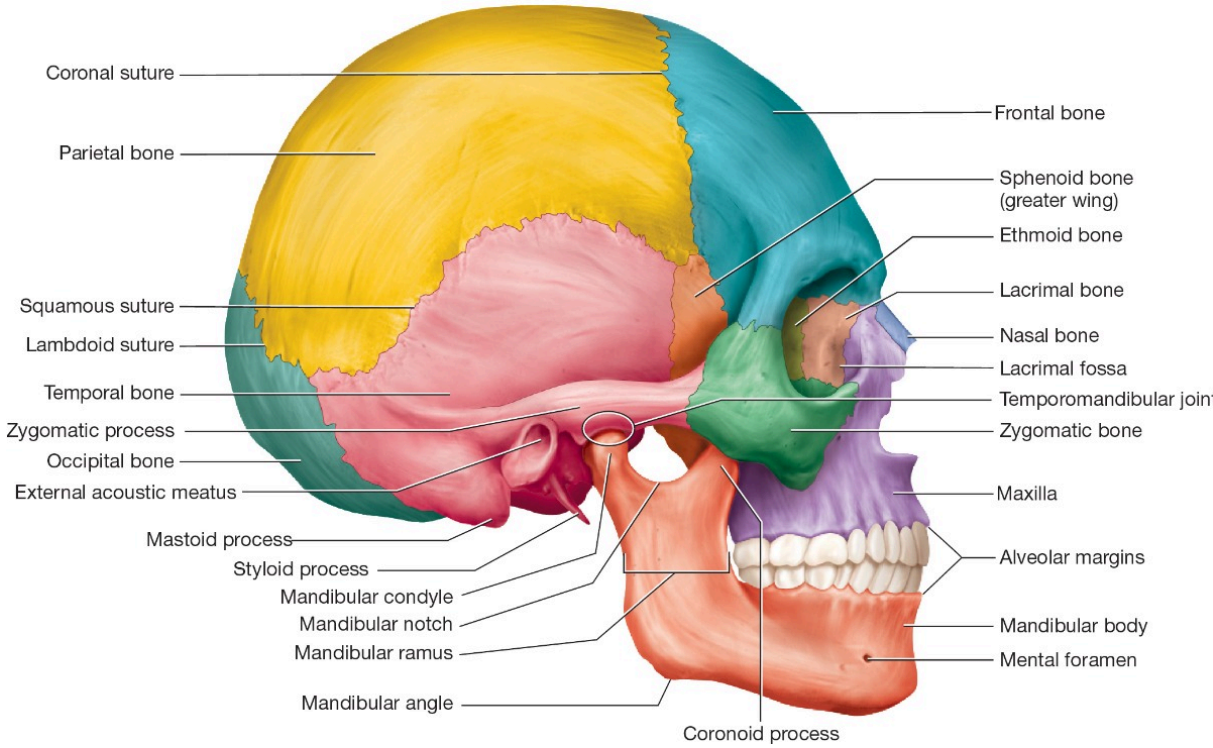


Figure 8.3 Lateral view of the skull.

Facial Bones

1. Notice that just lateral to the mental foramina, the mandibular rami turn superiorly at the **mandibular angle**. Their superior ends have two processes separated by the **mandibular notch**: an anterior process called the **coronoid process** (*coron-* = "hooked") and a posterior process called the **mandibular condyle**. The mandibular condyle fits into the mandibular fossa in the temporal bone to form the **temporomandibular joint**, or **TMJ**.
2. The maxillae, zygomatic bones, and nasal bones are visible in this view. The lacrimal bones are best seen from this view, where you can see the inferior indentation called the *lacrimal fossa*.

3. The inferior nasal conchae, vomer, and palatine bones are not visible here.

Internal View (Figure 8.6)

Cranial Bones

1. In the frontal bone, we can see the indentation known as the anterior cranial fossa.

2. In the middle of the frontal bone is the superior surface of the ethmoid bone, which is called the **cribriform plate** (KRIB-rih-form; *cribrum* = "sieve"). The cribriform plate forms the roof of the nasal cavity and part of the base of the cranial cavity. It has small holes, called **olfactory foramina** (*olfact-* = "smell"), through which olfactory nerves pass. It also has a superior projection, called the **crista galli** (KRIS-tah GAL-ee; "rooster's comb"), to which certain membranes around the brain attach.

3. Posterior to the frontal bone is the sphenoid bone. Centrally, the sphenoid bone consists of the **body**, the superior surface of which features two canals called the **optic canals** (*opt-* = "vision") through which the nerves for vision pass. Also in the body we find a saddle like formation called the **sella turcica** (SELL-uh TUR-sih-kuh; "Turk's saddle") that houses the pituitary gland. Extending from the body are three sets of "wings":

a. The small **lesser wings** are the superior and most anterior set.

b. The larger **greater wings** are the inferior and posterior set. As you can see in the figure, each greater wing has three prominent sets of foramina: the anterior **foramen rotundum**, the middle **foramen ovale** (oh-VAL-ay), and the posterior **foramen spinosum** (a mnemonic that might help you

remember this is “Rigatoni Over Spaghetti”). On the anterior side, the greater wings form the posterior portion of the orbit.

c. The inferior set of wings, the *pterygoid processes,* are discussed with the inferior view.

4. Internally, the temporal bone is shaped like a mountain ridge and is accordingly called the **petrous region** (*petr-* = “rocky”). Within this region, we find the middle cranial fossa. We also find several openings, including the **jugular foramen** (*jugular* = “neck”) and the **internal acoustic meatus**.

5. The posterior bone is the occipital bone. We can see its most conspicuous feature in its base—a large hole called the **foramen magnum** (“large hole”) through which the spinal cord passes. In addition, it features the posterior cranial fossa.

6. The parietal bones are visible on the lateral walls of the skull in this view.

Facial Bones

There are no facial bones visible in the internal view of the skull.

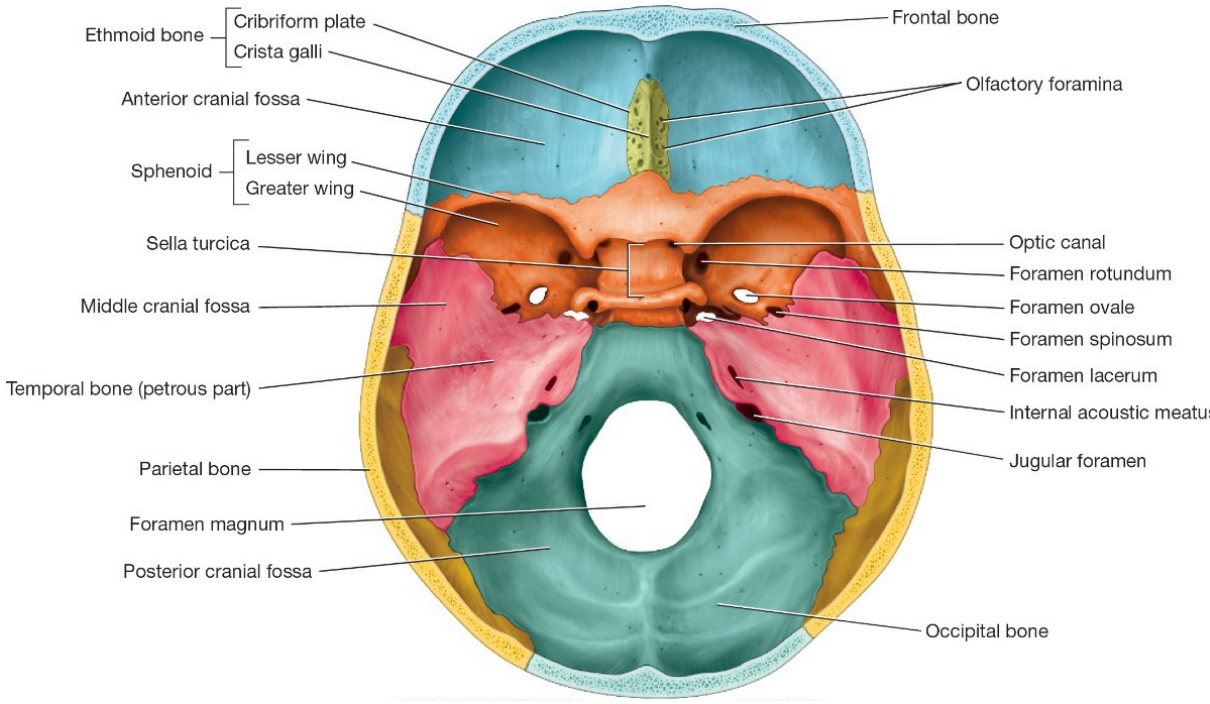


Figure 8.6 Internal view of the cranial base (calvaria removed).

Inferior View (Figure 8.7)

Cranial Bones

1. The frontal and ethmoid bones are not visible in this view. Only tiny slivers of the parietal bones are visible near the lateral edges of the skull.
2. Here we can see two sets of the sphenoid bone's wings. Laterally are its greater wings. Medially are the inferior set of wings, which consists of the narrow **pterygoid processes** (TEHR-ih-goyd; *ptery-* = "wing"). The pterygoid processes form part of the posterior walls of the nasal and oral cavities.
3. In this view, we can see many features of the temporal bone's tympanic and mastoid regions, including the medial and anterior **foramen lacerum** (LAS-er-um), the middle **carotid canal**, and the posterior jugular foramen. Also featured are the mandibular fossa, styloid process, and mastoid process.

4. From the inferior view, we can see many important features of the occipital bone. Anterior and lateral to the foramen magnum are the two **occipital condyles**, which form a joint with the first cervical vertebra. Also visible are two prominent horizontal ridges on the occipital bone's posterior surface: the **superior** and **inferior nuchal lines** (NOO-kuhl). In the middle of the superior nuchal line, we find a bump called the **external occipital protuberance**.

Facial Bones

1. The mandible, lacrimal bones, nasal bones, and inferior nasal conchae are not visible in the inferior view.
2. In this view, you can see that the maxillae also form the anterior portion of the *hard palate* via their **palatine processes**. At the anterior edge of the palatine processes, where the two maxillae meet, is the **incisive fossa**.
3. Posterior to the palatine processes are the palatine bones, which make up the posterior quarter of the hard palate.
4. Along the midline, between the pterygoid processes of the sphenoid bones, the vomer is visible.
5. Just lateral to the maxillae, you can see the zygomatic bones forming part of the zygomatic arch.

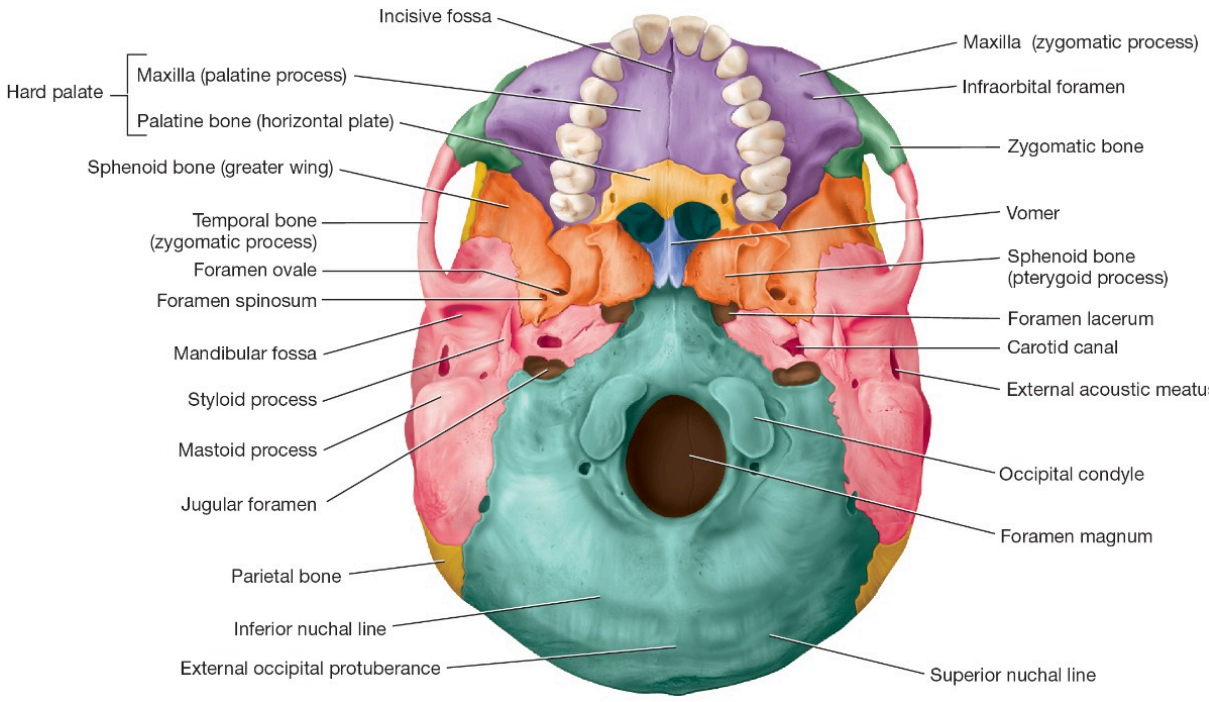


Figure 8.7 Inferior view of the skull (mandible removed).

Midsagittal Section (Figure 8.8)

Cranial Bones

1. In the frontal bone, notice the hollow **frontal sinuses**, which are part of the **paranasal sinuses** (*para-* = "next to"), a group of bony cavities that surround the nasal cavity.
2. You can see three of the sutures where the parietal bones articulate with other cranial bones: the lambdoid suture, the squamous suture, and the coronal suture. They also articulate with one another at the *sagittal suture*, which is visible in later figures (see Figures 8.7 and 8.8).
3. This view shows the petrous region of the temporal bone, and you can see clearly the internal acoustic meatus. In addition, the styloid process is visible.
4. Two features of the occipital bone to note in this view are the foramen magnum and the external occipital protuberance.

5. In the body of the sphenoid bone, notice the second paranasal sinus, the **sphenoid sinus**, which resides deep to the sella turcica. This is also a good view from which to appreciate the sphenoid bone's inferior set of wings—the narrow pterygoid processes.
6. This view shows the ethmoid bone's **perpendicular plate**, which forms the superior part of the bony nasal septum. You can also see its crista galli protruding into the cranial cavity.

Facial Bones

1. The body and ramus of the mandible are visible in this view, as is the mandibular condyle.
2. The maxilla and palatine bones can be seen making up the hard palate.
3. This section has the best view of the vomer—you can see how it makes up the inferior portion of the nasal septum.
4. A portion of one of the nasal bones is visible superficial to the perpendicular plate of the ethmoid bone.
5. The zygomatic bones, lacrimal bones, and inferior nasal conchae are not visible in this figure.

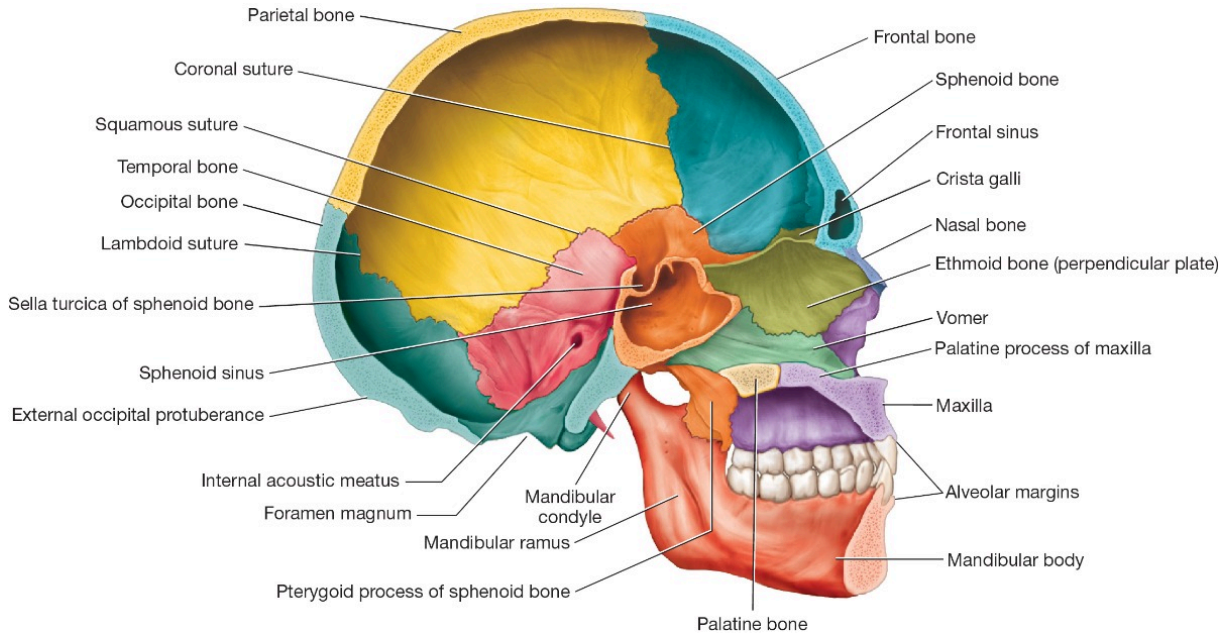


Figure 8.8 Midsagittal section of the skull.

Posterior View and Superior View (Figures 8.9 and 8.10)

Cranial Bones

1. This is the best view for seeing the shape of the occipital bone and the way it articulates with the parietal bones. You can also see the superior and inferior nuchal lines and the external occipital protuberance. On its inferior side is the foramen magnum.
2. From the posterior view, you can appreciate the size and shape of the temporal bones' mastoid processes and styloid processes.
3. In the superior view, the frontal bone is visible on the anterior end, and the occipital bone is visible on the posterior end.
4. In both views, we can see the cranial sutures. Recall that the parietal bones articulate with one another and many other cranial bones at sutures. In **Figures 8.9** and **8.10**, you can see that they meet one another at the **sagittal suture**. You can also see their junctions with the occipital bone at the lambdoid suture (**Figures 8.9** and **8.10**); with the temporal bones at the squamous sutures (**Figure 8.9**); and with the frontal bone at the coronal suture (**Figure 8.10**).

Facial Bones

Only the mandible and a small portion of the maxillae are visible from the posterior view; no facial bones are visible from the superior view.

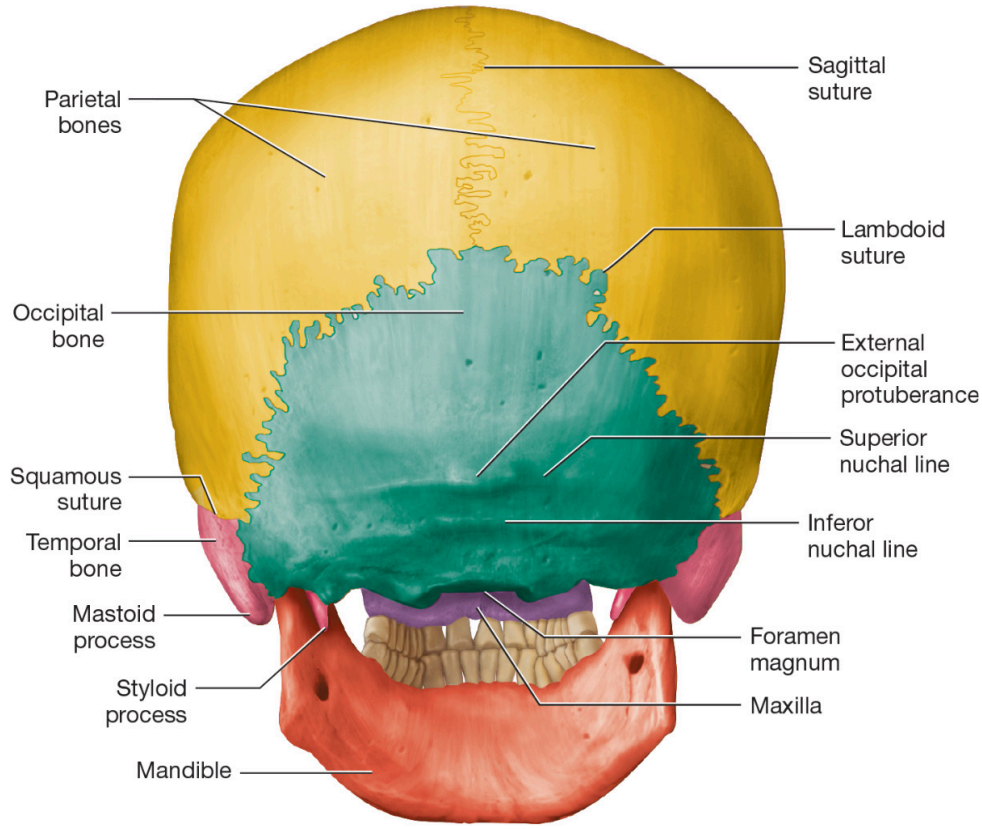


Figure 8.9 Posterior view of the skull.

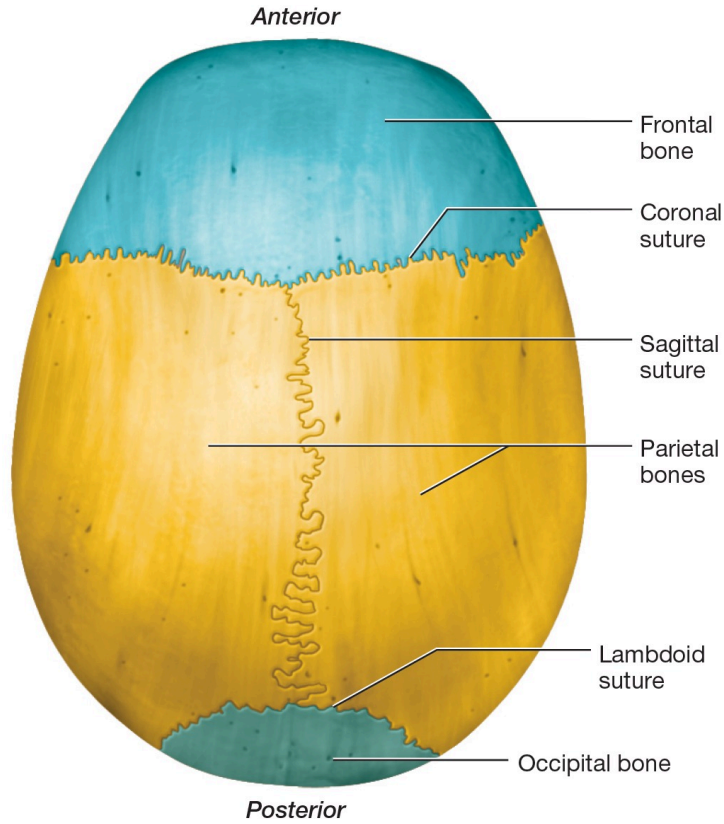


Figure 8.10 Superior view of the calvaria.

View of the Ethmoid Bone (Figure 8.11)

As the deepest cranial bone, the ethmoid bone is the most difficult one to see from standard views of the skull. For this reason, it is shown in a separate illustration, **Figure 8.11.**

1. Note the cribriform plate projecting from the ethmoid bone's superior - surface, with its superior projection, the crista galli.
2. The lateral masses form the *orbital plate* and the walls of the nasal cavity. Notice how internally the lateral masses contain numerous cavities—this is the third set of paranasal sinuses, the **ethmoid sinuses**. Extending medially from the lateral masses are two projections into the nasal cavity: the **superior nasal conchae** and **middle nasal conchae** (KAHN-kee). Note that the superior nasal conchae are quite small and are usually difficult to see.
3. You can see here how the middle perpendicular plate forms the superior part of the bony nasal septum.

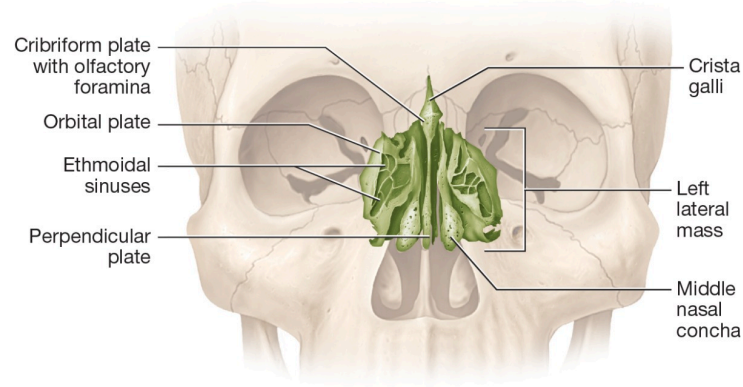


Figure 8.11 Ethmoid bone.

View of the Paranasal Sinuses (Figure 8.12)

So far, we have seen three sets of paranasal sinuses: the frontal, sphenoid, and ethmoid sinuses. There is a fourth set we have yet to discuss: the large **maxillary sinuses**, which are located within the maxillae. Air from the nasal

cavity enters the paranasal sinuses via small openings in the bones, and in the sinuses the air gets filtered, warmed, and humidified. The paranasal sinuses also decrease the weight of the skull and enhance voice resonance—which is evident from the change in your voice any time you have a sinus infection. **Figure 8.12** illustrates all four sinuses so you can see how they relate to one another.

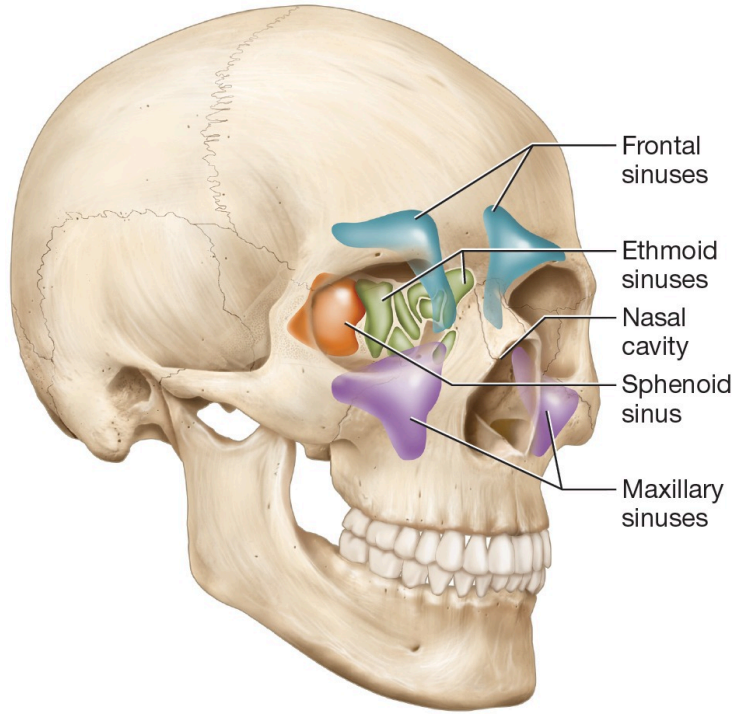


Figure 8.12 Paranasal sinuses.

Fetal Skull (Figure 8.13)

The features we've been examining are found in an adult skull. However, as you can see in **Figure 8.13**, a fetal skull contains notable differences from an adult skull. In adults, the sutures are fused, but in the fetus, the sutures have not yet fused and are instead joined by fibrous membranes. This can be seen with the **frontal suture**, also known as the **metopic suture** (met-AHP-ik; *metop-* = "forehead"), which is where the two fetal frontal bones fuse. Where several sutures meet, we find large, membrane-covered areas called the **fontanels**, known to many as "soft spots." The two main fontanels are the **anterior fontanel**, where the sagittal and coronal sutures meet, and the **posterior fontanel**, where the sagittal and lambdoid sutures meet.

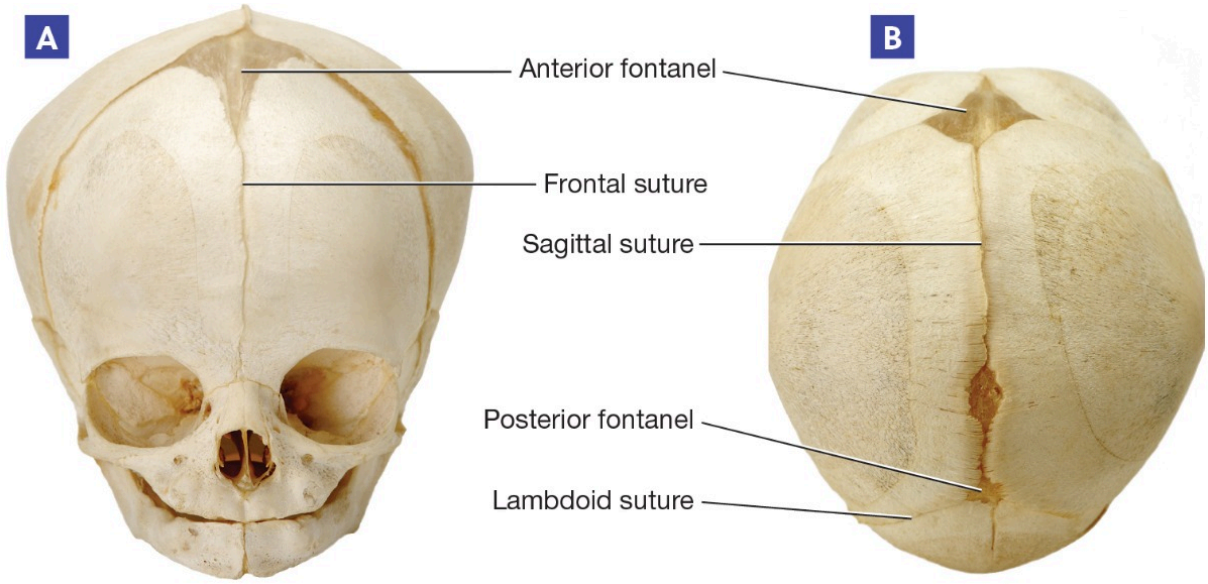


Figure 8.13 Fetal skull: (**A**) anterior view; (**B**) superior view.

Photo Gallery of the Skull

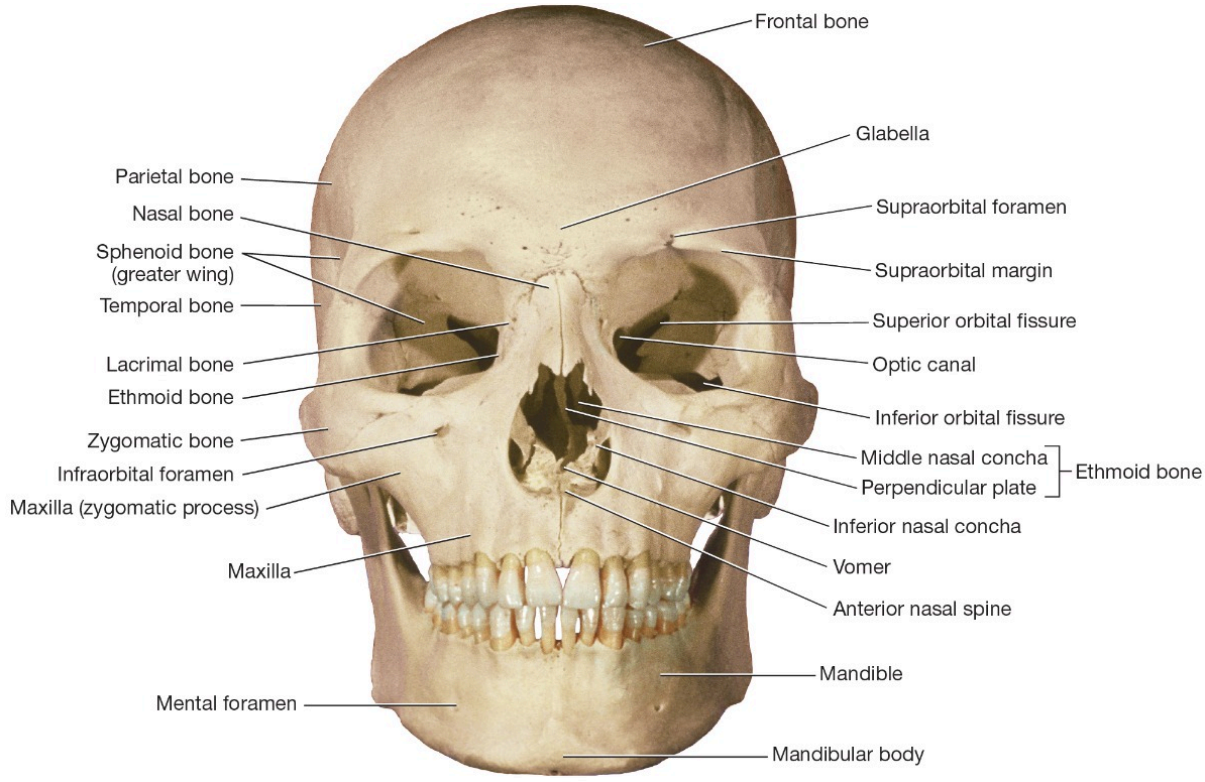


Figure 8.14 Anterior view of the skull, photograph.

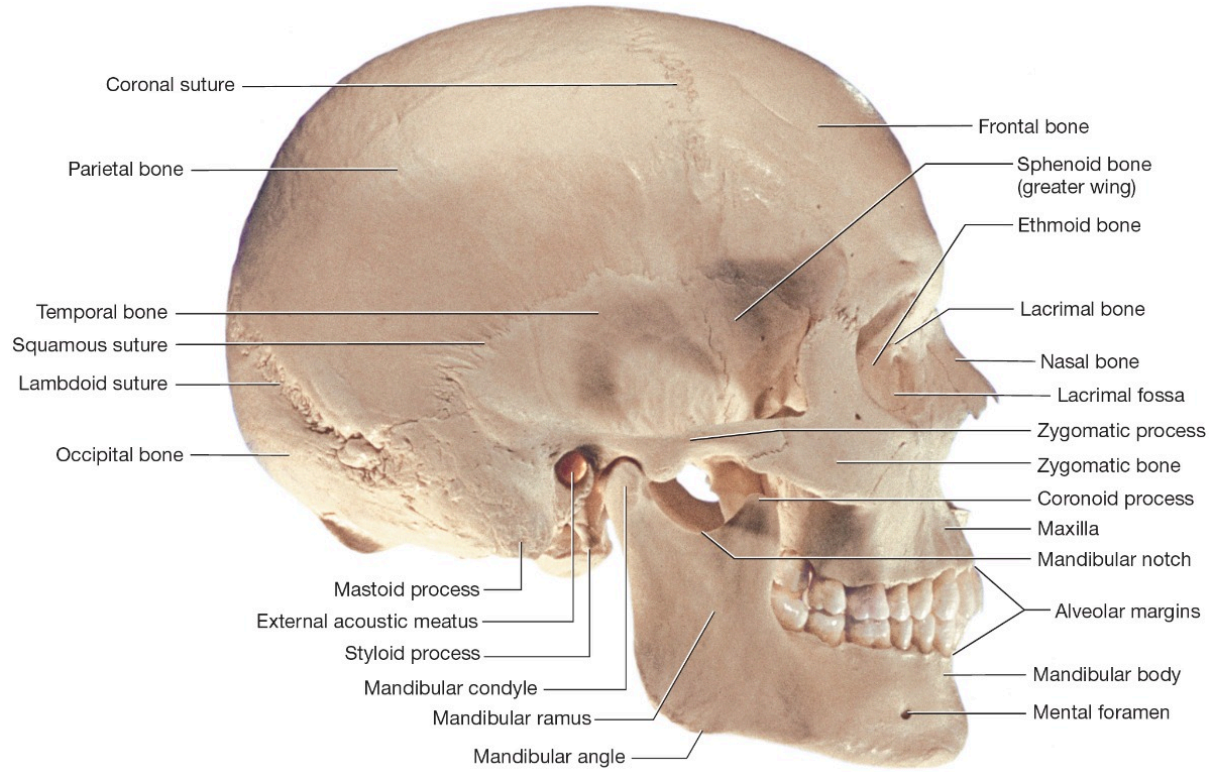


Figure 8.15 Lateral view of the skull, photograph.

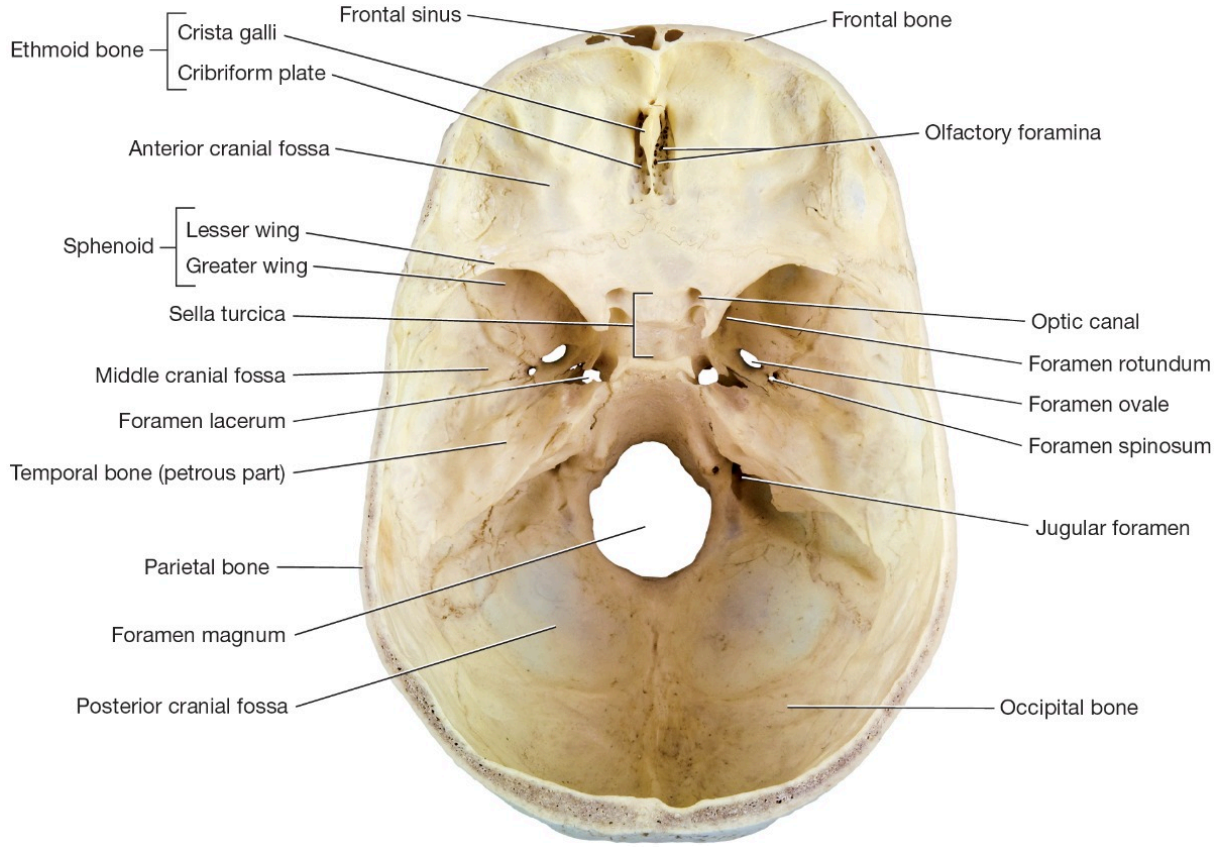


Figure 8.16 Internal view of the cranial base (calvaria removed), photograph.

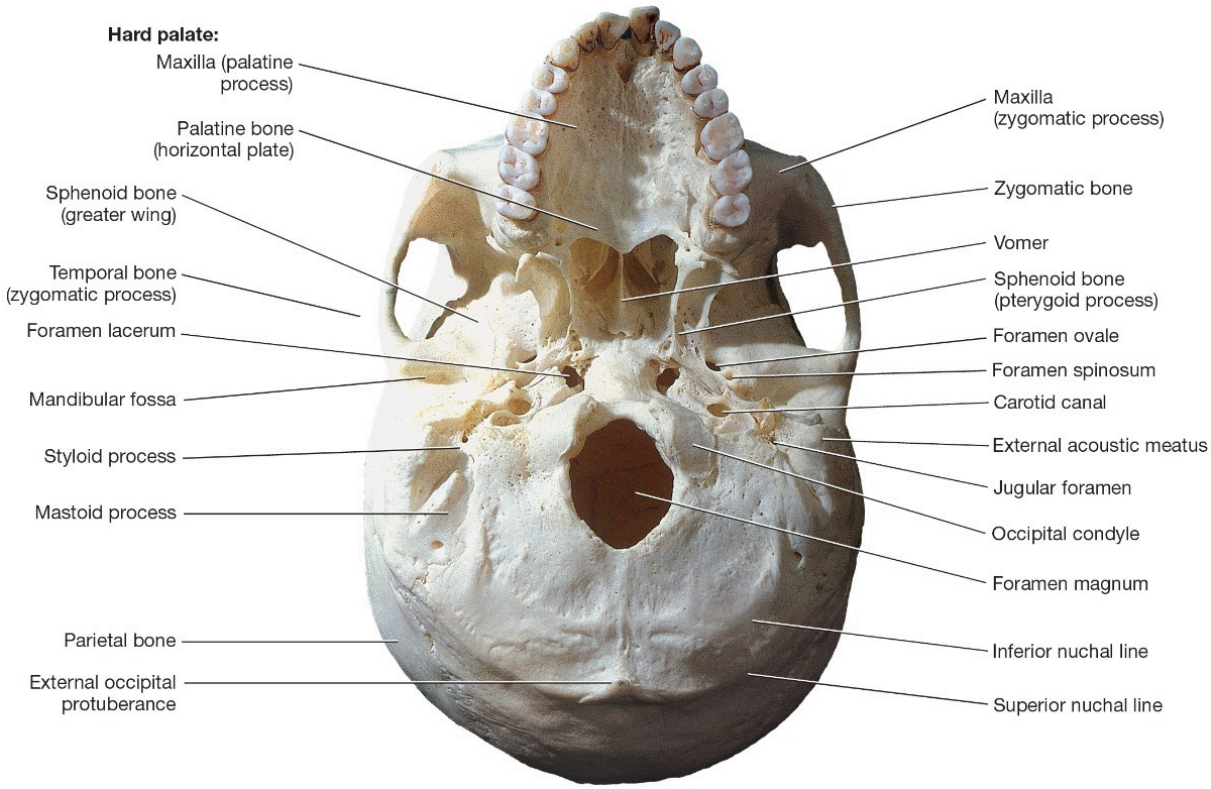


Figure 8.17 Inferior view of the skull (mandible removed), photograph.

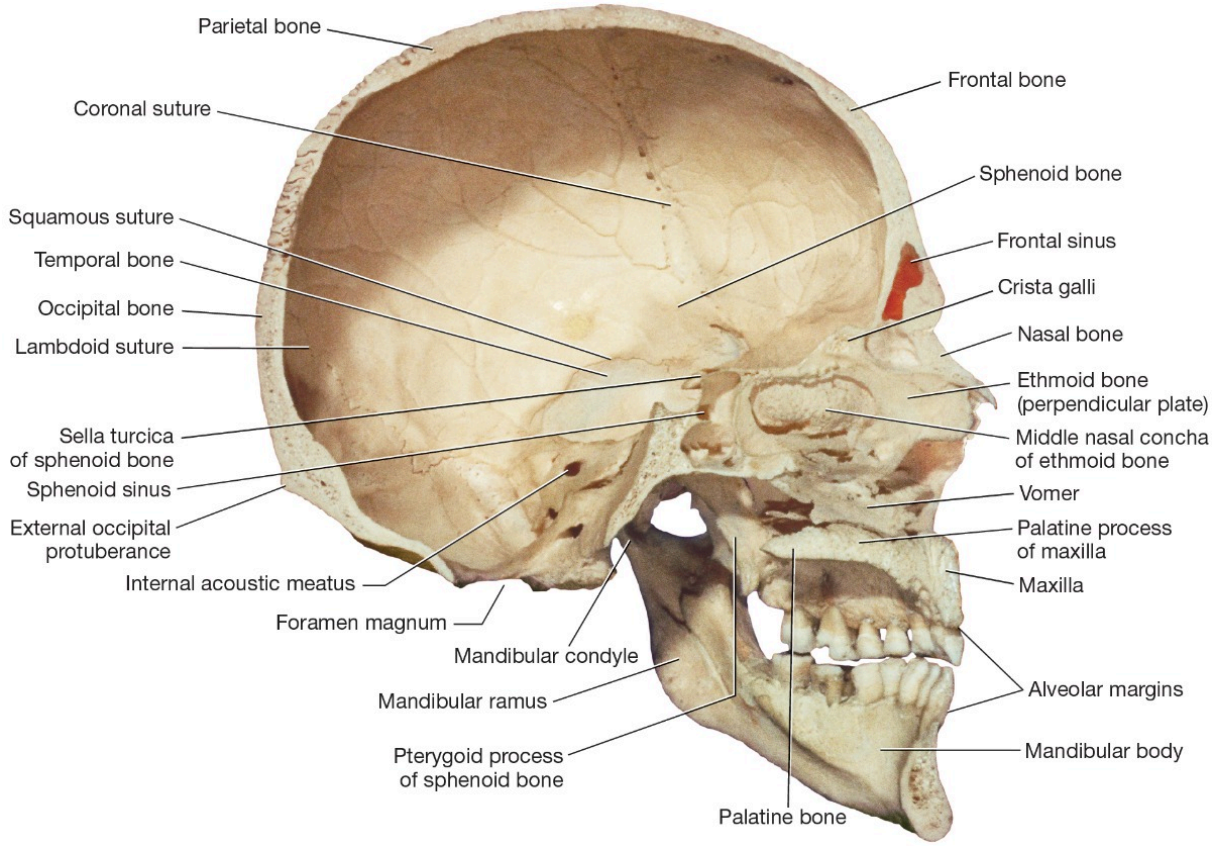


Figure 8.18 Midsagittal section of the skull, photograph.

Procedure 1: Cooperative Learning for the Skull

This exercise takes an approach called *cooperative learning*, in which you work with your lab partners to teach one another the bones and bone markings. The process may seem a bit confusing at first, but by the end of the first couple of rotations, it should move more quickly and smoothly.

1. Assemble into teams of a minimum of four students; five is optimum.

2. Distribute a skull to each member of the team, and assign each student one of the following five groups of bones and bone markings (the specific structures of each bone are listed below and on the following page):

- *Group A*: Calvaria, base, and structures of the frontal bone and parietal bones.
- *Group B*: Temporal bone and occipital bone structures.
- *Group C*: Ethmoid bone and sphenoid bone structures.
- *Group D*: Mandible and maxillary bone structures.
- *Group E*: Remainder of the facial bones, orbit, sutural bones, and anterior and posterior fontanels.

3. Spend approximately 3 minutes learning the assigned structures on your own. Explore each structure thoroughly. If there are canals and foramina, pass a pipe cleaner through the holes to see where they go.

4. Spend about 1 to 2 minutes teaching your assigned structures to your team. Then have each team member spend approximately 1 to 2 minutes teaching the team his or her assigned structures.

5. Rotate the assigned structures clockwise so each member has a new set of structures to learn and teach (the student who was assigned Group A will take Group B, and so on).

6. Repeat Steps 3 and 4, and then rotate the assigned structures again. Continue the process (it begins to speed up significantly at this point), repeating until each team member has taught each group of structures once.

By the end of this activity, each team member will have learned and presented each group of structures.

The following is a list of bones and bone markings of the skull that will be covered in this exercise.

Cranial Bones

Group A Structures

1. Calvaria
2. Base of cranial cavity
    a. Anterior cranial fossa
    b. Middle cranial fossa
    c. Posterior cranial fossa

3. Frontal bone
    a. Glabella
    b. Supraorbital foramen
    c. Frontal sinuses

4.Parietal bones
    a. Sagittal suture
    b. Coronal suture
    c. Lambdoid suture
    d. Squamous suture

Group B Structures

1. Temporal bones

- **a.** Zygomatic process
- **b.** Mandibular fossa
- **c.** External acoustic (auditory) meatus
- **d.** Styloid process
- **e.** Mastoid process with mastoid air cells
- **f.** Foramen lacerum
- **g.** Carotid canal
- **h.** Jugular foramen
- **i.** Internal acoustic meatus

2. Occipital bone

- **a.** Foramen magnum
- **b.** Occipital condyles
- **c.** Superior nuchal line
- **d.** Inferior nuchal line
- **e.** External occipital protuberance

Group C Structures

1. Sphenoid bone

- **a.** Body
- **b.** Optic canal
- **c.** Sphenoid sinus
- **d.** Sella turcica
- **e.** Lesser wings
- **f.** Greater wings
- **g.** Superior orbital fissure
- **h.** Foramen rotundum
- **i.** Foramen ovale
- **j.** Foramen spinosum
- **k.** Pterygoid processes

2. Ethmoid bone

- **a.** Cribriform plate
- **b.** Olfactory foramina
- **c.** Crista galli
- **d.** Lateral masses
- **e.** Ethmoid sinuses
- **f.** Superior and middle nasal conchae
- **g.** Perpendicular plate

Facial Bones and Other Structures

Group D Structures

1. Mandible

- **a.** Body of mandible
- **b.** Mandibular ramus

c. Alveolar margin

- **d.** Mental foramen
- **e.** Mandibular angle
- **f.** Mandibular notch
- **g.** Coronoid process
- **h.** Mandibular condyle
- **i.** Temporomandibular joint

2. Maxillae

- **a.** Inferior orbital fissure
- **b.** Infraorbital foramen
- **c.** Palatine processes
- **d.** Maxillary sinuses
- **e.** Zygomatic processes
- **f.** Alveolar margin

Group E Structures

1. Lacrimal bones and lacrimal fossa
2. Nasal bones
3. Vomer
4. Inferior nasal conchae
5. Palatine bones
6. Zygomatic bones and zygomatic arch
7. Orbit
8. Nasal cavity
9. Frontal (metopic) suture
10. Anterior fontanel
11. Posterior fontanel

Exercise Question 8.1

Your instructor may wish to omit certain structures included in these lists or add structures not included in the lists. List any additional structures in the space provided:

Your Answer

No answer submitted

Exercise 8-2 Remainder of the Axial Skeleton

Another key component of the axial skeleton is the **vertebral column**, which consists of 24 unfused vertebrae, the five fused vertebrae of the sacrum (SAY-krum), and the coccyx (KAHX-iks). If you view the vertebral column from the lateral side, as in **Figure 8.19**, you can see that there are spaces between the vertebrae called **intervertebral foramina** (*inter-* = “within” or “between”). In

addition, the vertebral column has four curvatures: the concave **cervical** (*cerv-* = "neck") and **lumbar curvatures** (*lumb-* = "loin"), and the convex **thoracic** (*thorac-* = "chest") and **sacral curvatures**. The cervical and lumbar curvatures are particularly important to our ability to walk upright.

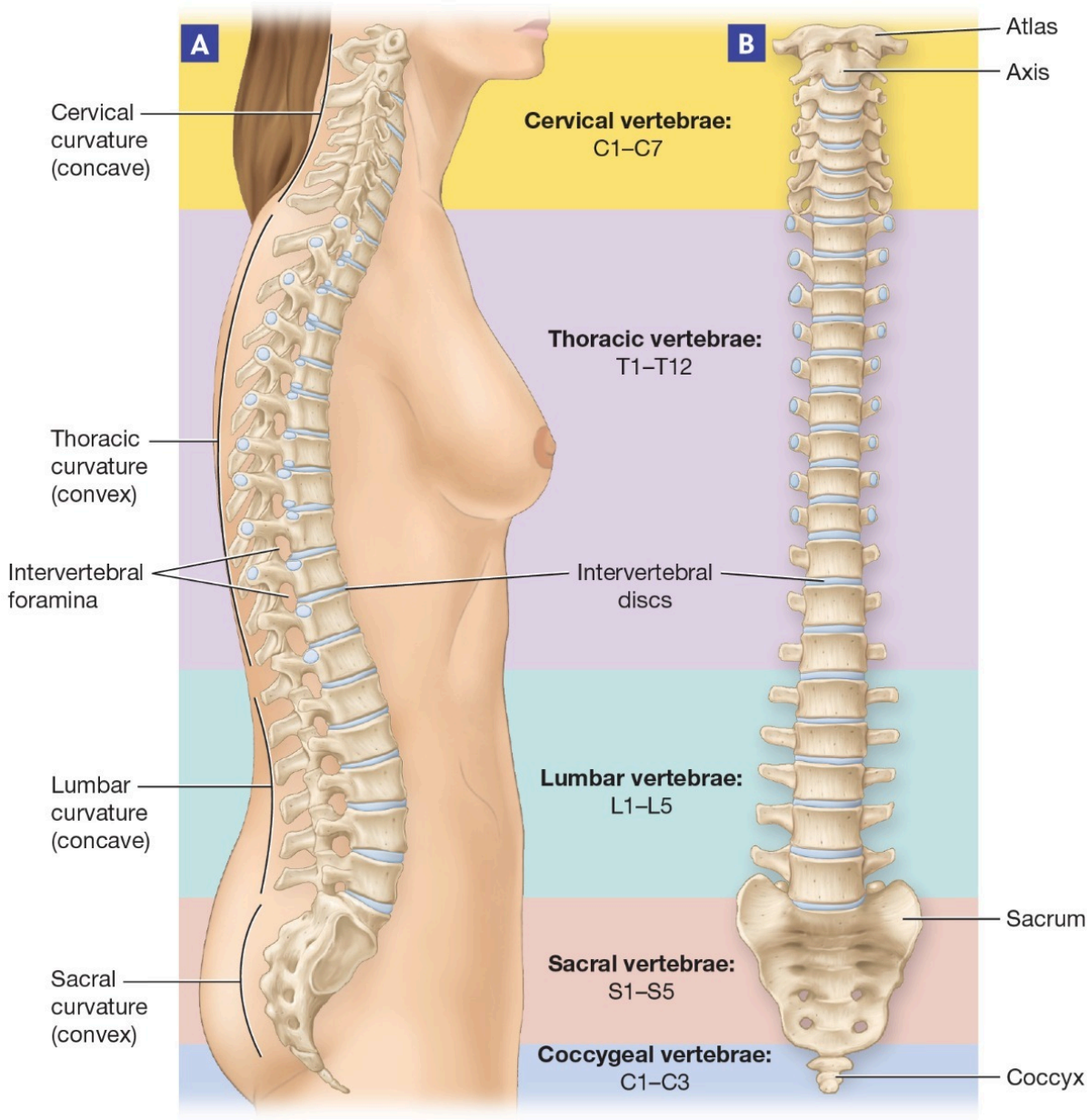


Figure 8.19 Vertebral column: (**A**) lateral view; (**B**) anterior view.

The 24 vertebrae consist of 7 cervical vertebrae, 12 thoracic vertebrae, and 5 lumbar vertebrae (remember this as "breakfast at 7, lunch at 12, dinner at 5"). Nearly all vertebrae share certain general features, which are visible in the typical cervical vertebra shown in **Figures 8.20A** and **8.20B**. These features include a posterior **spinous process**, two lateral **transverse processes**, a central **vertebral foramen**, an anterior **vertebral body**, two **superior**

articular processes (*artic-* = "joint") and **facets** (fuh-SETZ) that articulate with the vertebra above, and two inferior articular processes and facets that articulate with the vertebra below. In between each vertebral body is a fibrocartilage pad called an **intervertebral disc** that absorbs shock as the vertebral column moves. Extending from the vertebral body to the transverse processes on both sides are two short extensions known as **pedicles** (PED-ih-kullz), which—along with two posterior extensions, the **laminae** (LAM-ih-nee)—enclose the vertebral foramen.

The basic properties of each region of the vertebral column are as follows:

1. The seven **cervical vertebrae** are located in the neck (**Figure 8.20A-D**). All cervical vertebrae have holes in their transverse processes called **transverse foramina**. These foramina permit the passage of blood vessels called the *vertebral artery* and *vein*.

In addition, the spinous processes of cervical vertebrae are often forked. Two cervical vertebrae are named differently than the others because of their unique features:

a. Atlas (C1): The **atlas** is the first cervical vertebra, and it articulates with the occipital condyles of the occipital bone. It is easily identified because it has a large vertebral foramen, no body, and no spinous process (**Figure 8.20C**).

b. Axis (C2): The **axis** is the second cervical vertebra. It is also easily identified by a superior projection called the **dens** (or the **odontoid process**; **Figure 8.20D**). The dens fits up inside the atlas to form the *atlantoaxial joint*, which allows rotation of the head.

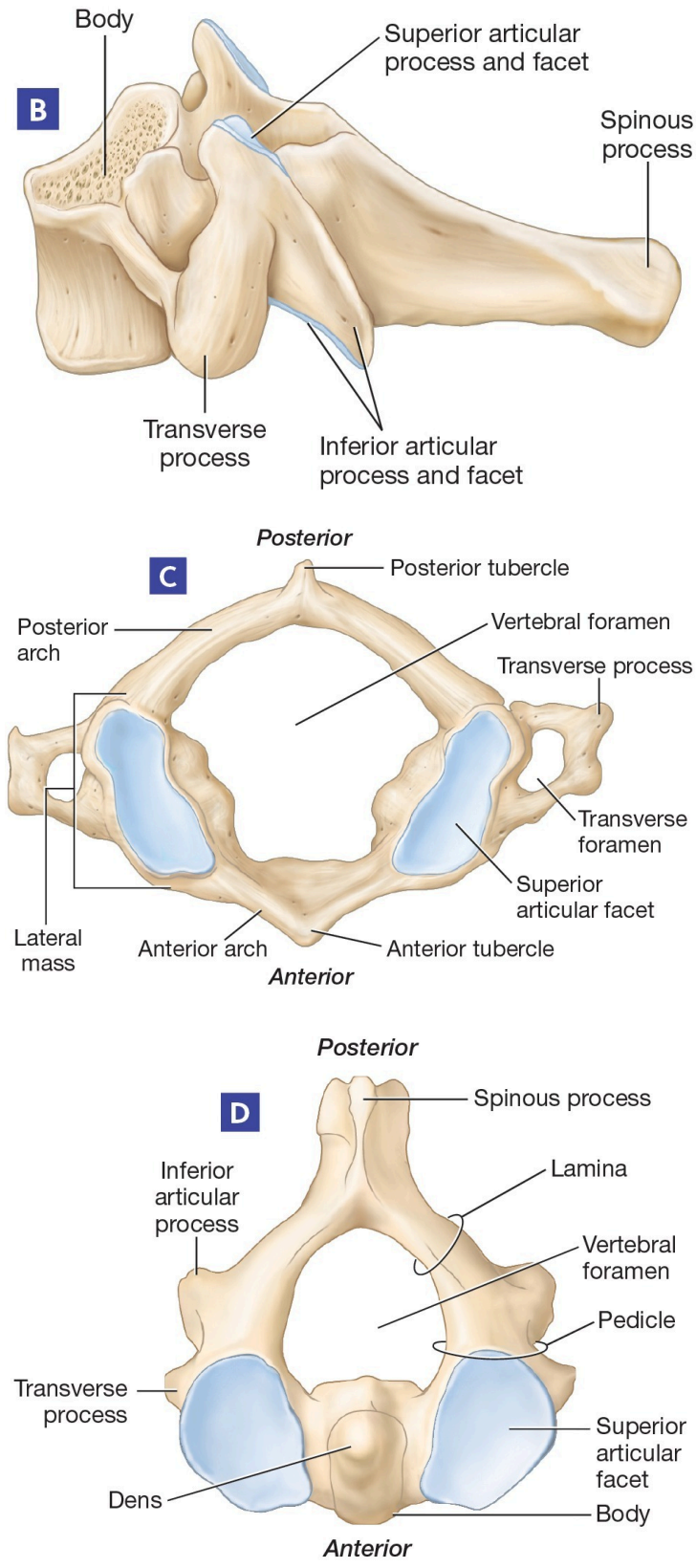


Figure 8.20 Cervical vertebrae: **(A)** typical cervical vertebra, superior view; **(B)** typical cervical vertebra, lateral view; **(C)** atlas (C1) **(D)** axis (C2).

2. The 12 **thoracic vertebrae** each articulate with a pair of ribs, and they share the following common features **(Figure 8.21)**:

a. The spinous processes are thin and point inferiorly.

b. They lack transverse foramina.

c. Most have two *costal facets* that articulate with the ribs (there are 12 pairs of ribs). They also have *transverse costal facets* on their transverse processes that articulate with a part of the rib called the *tubercle*.

d. All have approximately triangular vertebral foramina.

e. If you look at a thoracic vertebra from the posterolateral side, it looks like a giraffe. Try it—it really does!

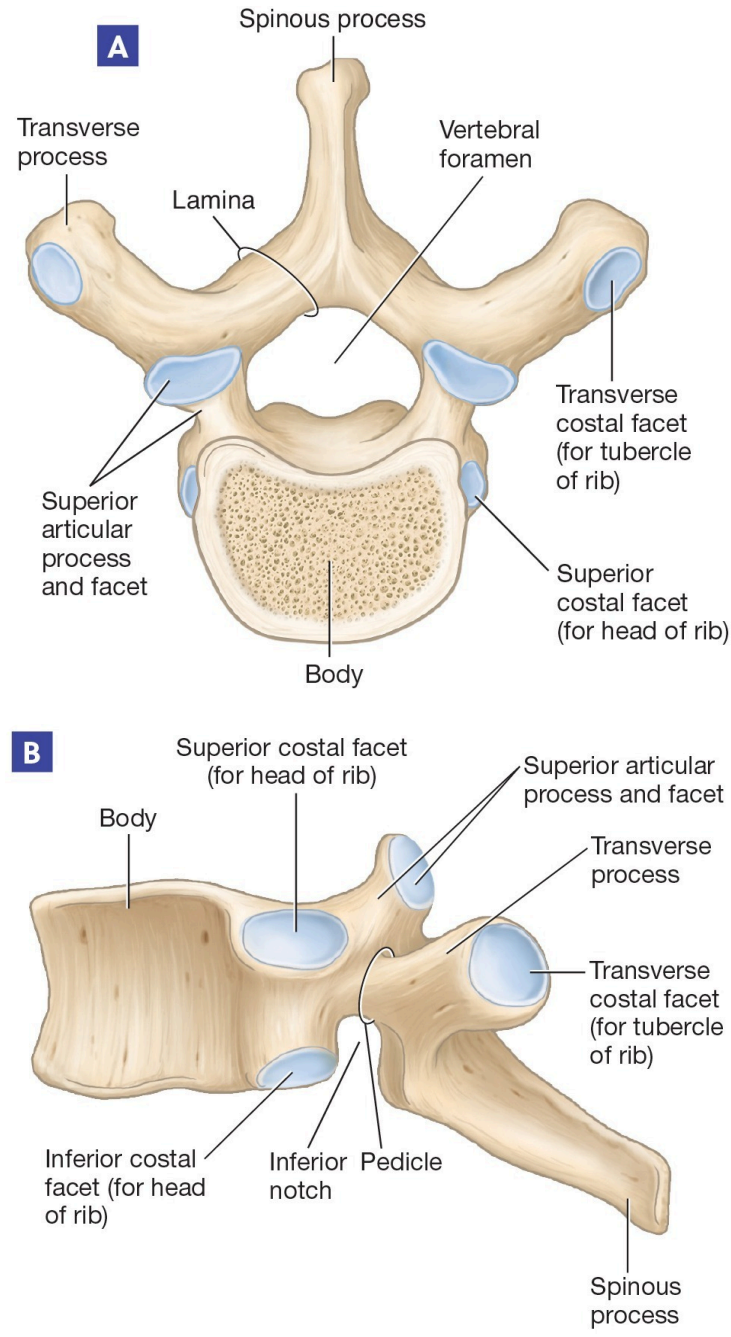


Figure 8.21 Thoracic vertebrae: **(A)** typical thoracic vertebra, superior view; (**B**) typical thoracic vertebra, lateral view.

3. The five large **lumbar vertebrae** share the following common features (**Figure 8.22**):

a. All have a large, block-like body.

b. The spinous processes are thick and point posteriorly.

c. They lack transverse foramina and costal facets.

d. If you look at a lumbar vertebra from the posterolateral side, it looks like a moose.

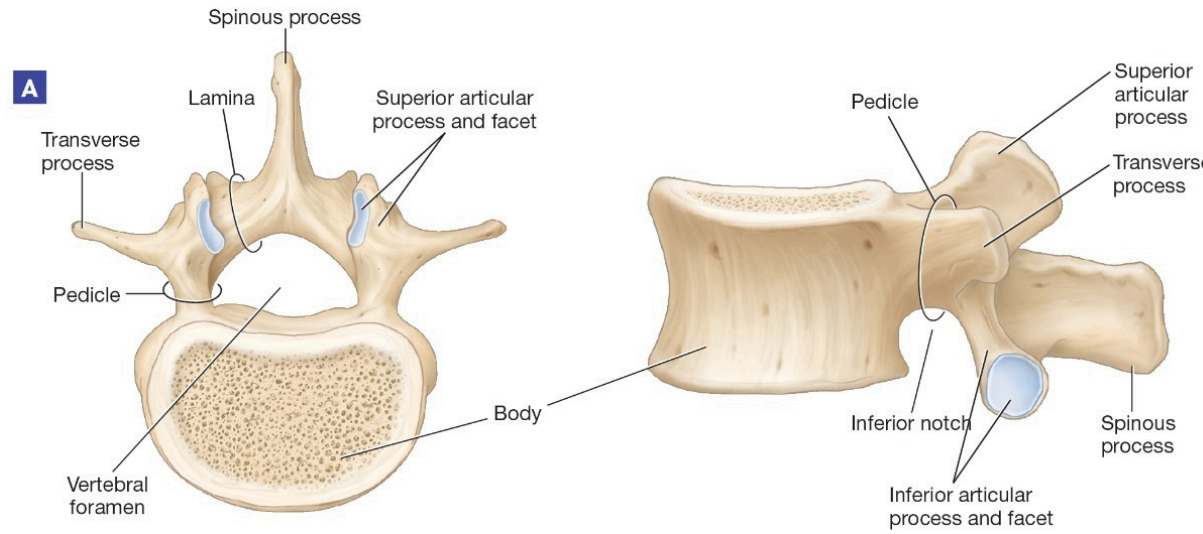


Figure 8.22 Lumbar vertebrae: **(A)** typical lumbar vertebra, superior view; **(B)** typical lumbar vertebra, lateral view.

4. The **sacrum** consists of five fused vertebrae (**Figure 8.23**). It's a fun note that the word "sacrum" means "sacred bone," reflecting the belief that it was the residence of the soul. On the sacrum's superior surface, we find the **sacral promontory**, which is the portion of the first sacral body that projects anteriorly into the pelvic cavity. Lateral to the sacral promontory are smooth regions known as the **alae** (AY-lee; *ala* = "wing"). Just posterior to the sacral promontory is the **sacral canal**, through which the inferior bundle of spinal nerve roots passes from the vertebral foramen of the fifth lumbar vertebra. You can see the fused spinous processes of the vertebrae on the sacrum's posterior side as the **median sacral crest**. Spinal nerves pass through holes called **anterior** and **posterior sacral foramina** that flank both sides of the sacral bodies. The lateral surfaces of the sacrum, known as the *auricular surfaces*, articulate with the hip bones to form the **sacroiliac joints** (say-kroh-ILL-ee-ak).

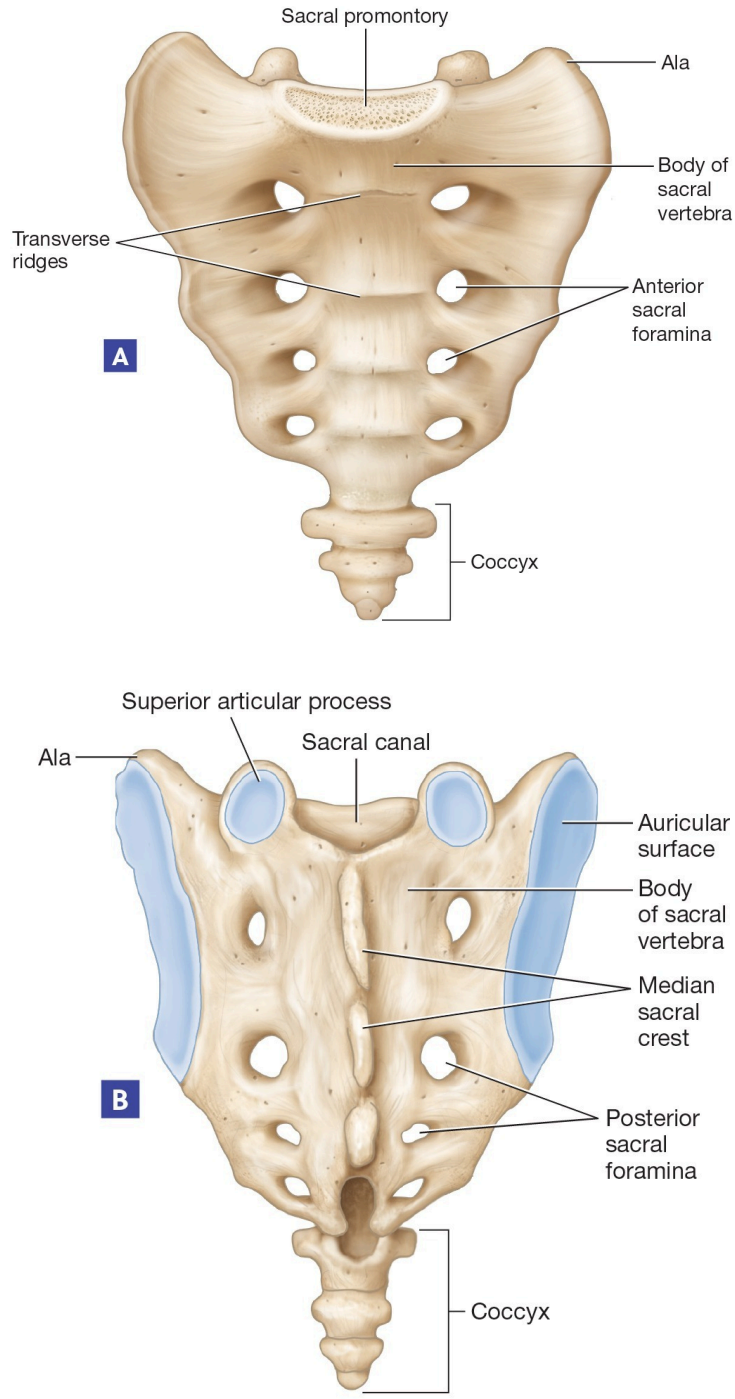


Figure 8.23 Sacrum: **(A)** anterior view; (**B**) posterior view.

5. The **coccyx** consists of three to five (average: four) small, fused vertebrae that articulate superiorly with the sacrum.

The remainder of the axial skeleton consists of the bones of the thoracic cavity (the sternum and the ribs; Figures **8.24** and **8.25**) and the hyoid bone in the neck (Figure 8.24). The **sternum** (*stern-* = "chest") is the central bone of the thorax. It is divided into three parts: the superior **manubrium** (mah-NOO-bree-um), the middle **body**, and the inferior **xiphoid process** (ZY-foyd; *xiphos* =

"sword"). It is the body of the sternum that you compress during the chest compressions of cardiopulmonary resuscitation (CPR), and the xiphoid process that you must take care not to break.

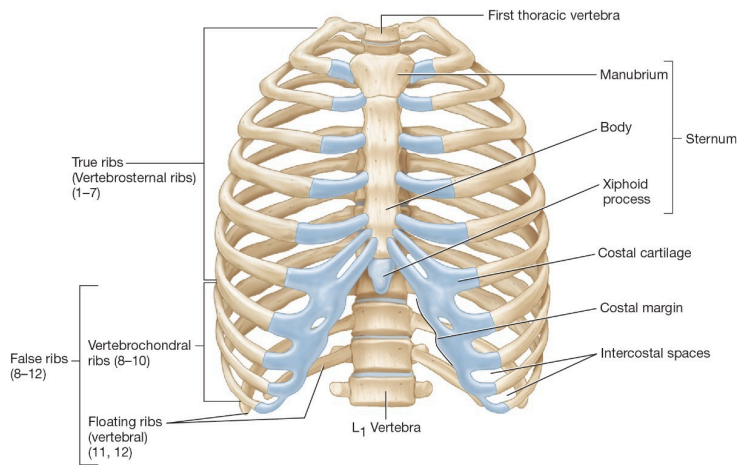


Figure 8.24 Thoracic cage.

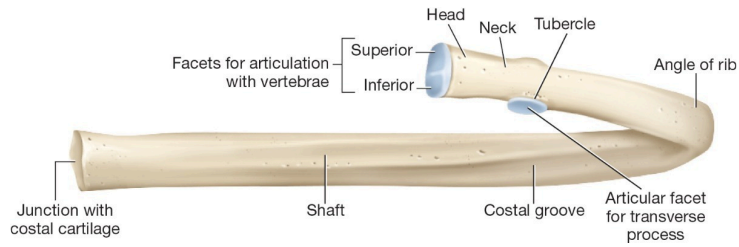


Figure 8.25 A rib, posterior view.



Figure 8.26 Hyoid bone.

The 12 pairs of **ribs** enclose the thoracic cavity and protect its vital organs (note that men and women have the same number of ribs). The spaces between the ribs are known as **intercostal spaces** (*cost-* = "rib"), and they contain muscles, blood vessels, nerves, and lymphatic vessels. Ribs articulate anteriorly with the sternum via **costal cartilage**. They are classified according to the structure of their costal cartilage: ribs 1–7 are considered **true ribs** or **vertebrosternal ribs**, because they attach directly to the sternum by their own costal cartilage, whereas ribs 8–12 are classified as **false ribs**, because they

lack this direct attachment to the sternum. However, ribs 8–10 do attach to the sternum via the cartilage of the true ribs and so they are called **vertebrochondral ribs.** Ribs 11–12 lack any attachment to the sternum, so they are often referred to as **floating ribs** or **vertebral ribs**. The medial border of the costal cartilage of the false ribs is known as the **costal margin**.

In **Figure 8.25**, you can see the rib's **superior** and **inferior facets** on the rib's posterior **head**. Here the rib straddles two thoracic vertebral bodies—the superior facet of the rib articulates with the inferior costal facet of the vertebra above, and the inferior facet of the rib articulates with the superior costal facet of the rib below. Lateral to the head, we find the **neck**, which contains a **tubercle** that articulates with a vertebra's transverse process. The rib curves around anteriorly at its **angle**, after which the main anterior portion becomes known as the **shaft**. On the shaft's interior surface is the **costal groove**, which is where blood vessels and nerves travel.

The **hyoid bone** (hy-OYD), shown in Figure 8.24, is located in the superior neck, where it is occasionally classified as a skull bone. Note, however, that it does not articulate with any skull bone, or any other bone, for that matter. It is held in place in the superior neck by muscles and ligaments, and it helps to form part of the framework for the larynx (LEHR-inks; voice box). It also serves as an attachment site for the muscles of the tongue and aids in swallowing. When a person is choked manually, the hyoid bone is often broken.

The bones and bone markings of the axial skeleton are listed with Exercise 8-3, where they will be included with another cooperative learning exercise.

Anatomy of the Appendicular Skeleton

The bones of the appendicular skeleton can be divided into those of the pectoral girdle, the upper limb, the pelvic girdle, and the lower limb. Let's look at each individually.

Pectoral Girdle

The **pectoral girdle** (pek-TOHR-uhl; *pect-* = chest) consists of the two bones that frame the shoulder: the scapula and the clavicle (**Figure 8.27**). The **scapula** (SKAP-yoo-luh; *scapul-* = "shoulder"), shown in **Figure 8.28**, is a roughly triangular-shaped bone with **superior**, **lateral**, and **medial borders**. The corners of the "triangle" are known as the **superior**, **lateral**, and **inferior angles**. The scapula's broad, flat surface is the **body**. On the anterior surface (**Figure 8.28A**), the body has a broad indentation called the **subscapular fossa** (*sub-* = "below"). On its lateral surface (**Figure 8.28B**) is a shallow depression called the **glenoid cavity** (*glen-* = "socket") that forms the shoulder joint with the humerus. From this side, you can also get a good view of its two superior projections: the anterior **coracoid process** (KOHR-ah-koyd) and the posterior **acromion** (ah-KROH-mee-ahn; *acrom-* = "highest"). The acromion joins with the lateral portion of the clavicle to form the **acromioclavicular** (ah-KROH-mee-oh-klah-VIK-yoo-lur; **AC**) joint. When we turn to the scapula's posterior surface (**Figure 8.28C**), we find a prominent ridge called the **spine** (you can feel this under your skin as your "shoulder blade"). Superior and inferior to the scapular spine are two depressions: the **supraspinous** and **infraspinous fossae**.

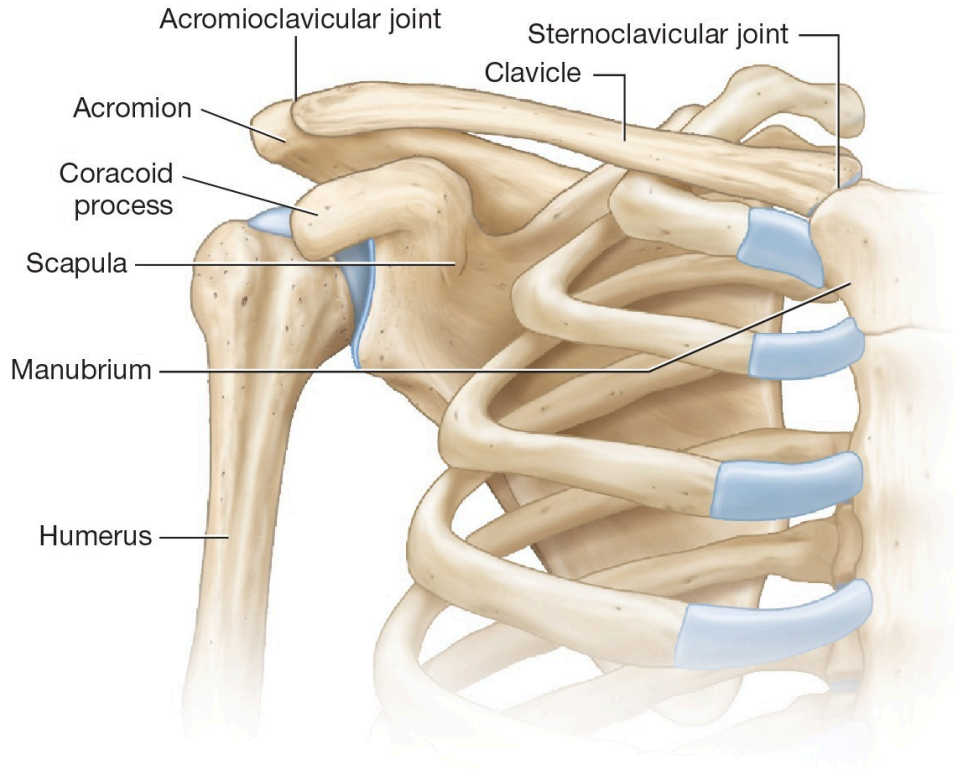


Figure 8.27 Pectoral girdle.

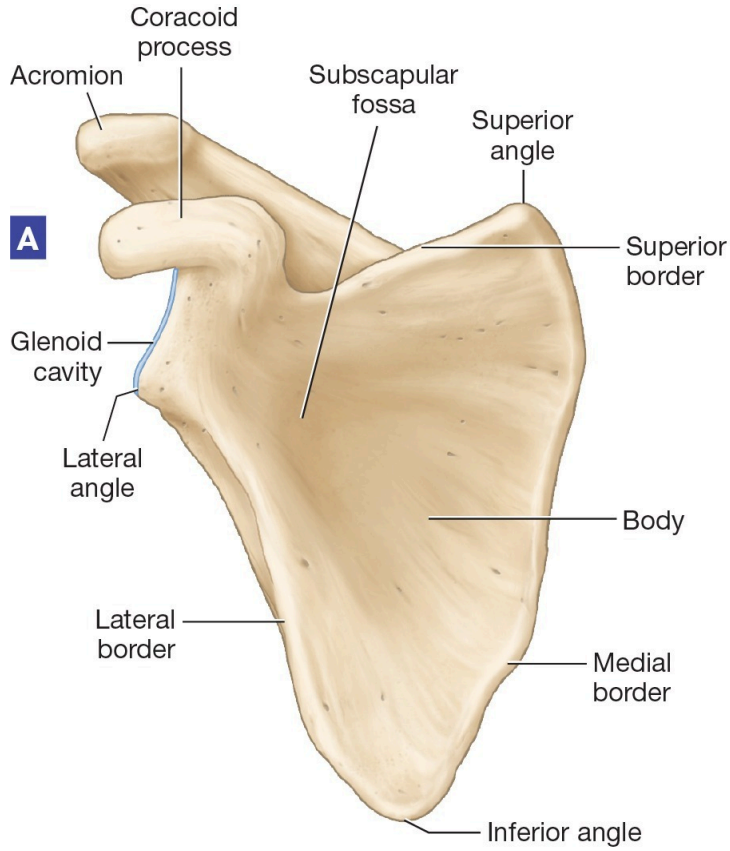


Figure 8.28 Right scapula: (**A**) anterior view; (*continues*)

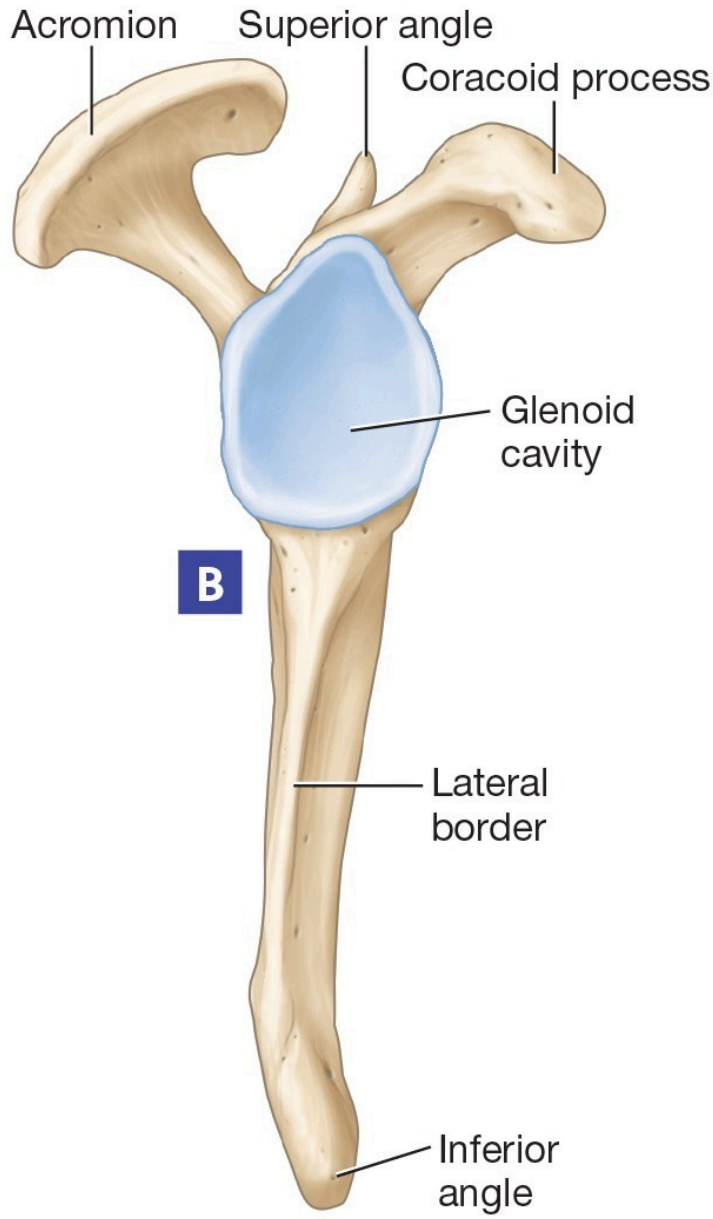


Figure 8.26: Right scapula: (*cont.*) (**B**) lateral view.

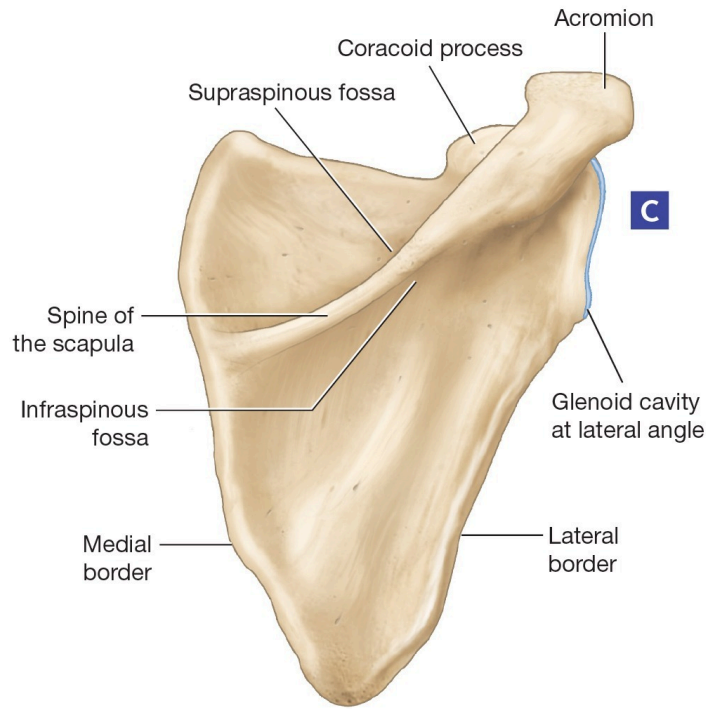


Figure 8.26: Right scapula: *(cont.)* (**C**) posterior view.

The **clavicle** is a small S-shaped bone that spans between the acromion of the scapula at its lateral acromial end and the manubrium of the sternum at its medial **sternal end**. Note its "S" shape is best appreciated from the superior view shown in **Figure 8.29**, as the clavicle appears nearly straight from an anterior view. Functionally, the clavicle acts somewhat like a brace, holding the upper limb in place away from the body.

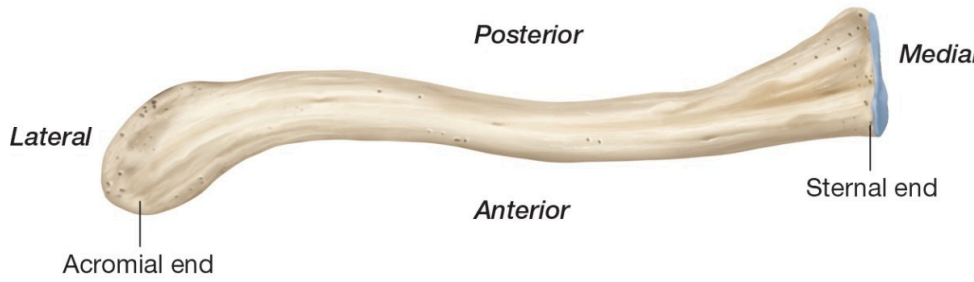


Figure 8.29 Clavicle, superior view.

Upper Limb

The **upper limb** consists of the arm, the forearm, the wrist, and the hand. The only bone within the arm is the **humerus** (HYOO-mur-uhs; *humerus* = "upper arm"), a long bone that forms the shoulder joint with the scapula at its

proximal end and the elbow joint with the ulna and radius at its distal end. The humerus has a number of features, including the following (**Figure 8.30**):

- At its proximal end is a rounded **head** that fits into the glenoid cavity. Just lateral to the head is the **greater tubercle**, separated from the smaller **lesser tubercle** by the **intertubercular sulcus** (in-ter-too-BUR-kyoo-lur SUL-kuhs).
- The diaphysis of the humerus features a projection called the **deltoid tuberosity**, where the deltoid muscle attaches. On the posterior diaphysis we find another feature—the **radial groove**—which is an indentation along which the radial nerve travels.
- At the humerus' distal end, we find two small projections called the **medial** and **lateral epicondyles**. Distal to these are two larger condyles: the medial **trochlea** (TROH-klee-uh), which is shaped like a spool of thread, and the lateral **capitulum** (kah-PIT-yoo-lum; *capit-* = "head"), which is ball-shaped. Just proximal to the trochlea are indentations in the humerus where the ulna and radius articulate—the anterior **coronoid** and **radial fossae** and the posterior **olecranon fossa** (oh-LEK-ruh-nahn; *olec-* = "elbow").

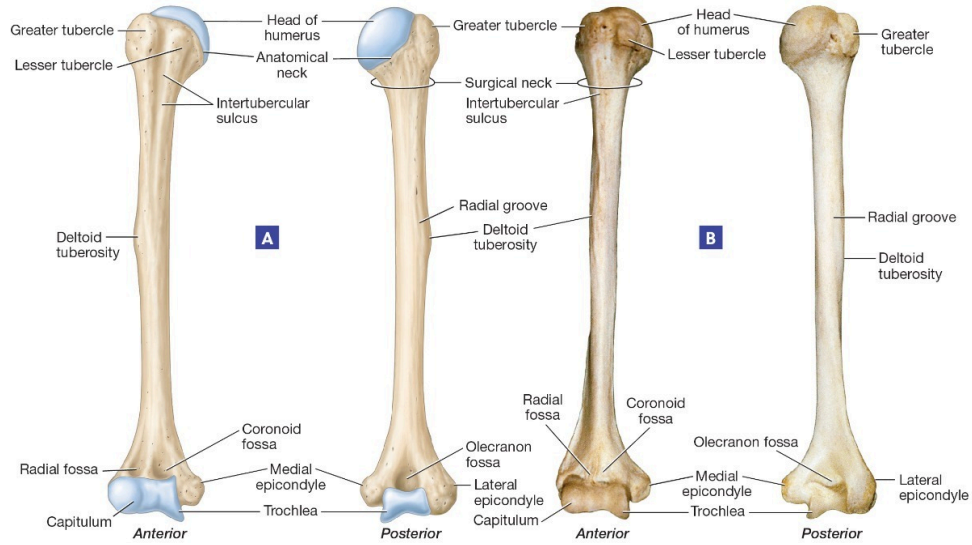


Figure 8.30 Right humerus: (**A**) illustration, anterior and posterior views; (**B**) photograph, anterior and posterior views.

The two forearm bones, shown in **Figure 8.31**, are the lateral **radius** (RAY-dee-uhs) and medial **ulna** (*ulna* = "elbow;" if you have a hard time remembering which is which, stand in anatomical position and take your radial pulse; the pulse is on the lateral side of the forearm, just like the radius). The ulna is wide proximally where it articulates with the humerus and thin distally where it articulates with the bones of the wrist. Its proximal end has two processes—the large, posterior **olecranon process** and the smaller, anterior **coronoid process**—separated by a deep curve called the **trochlear notch**.

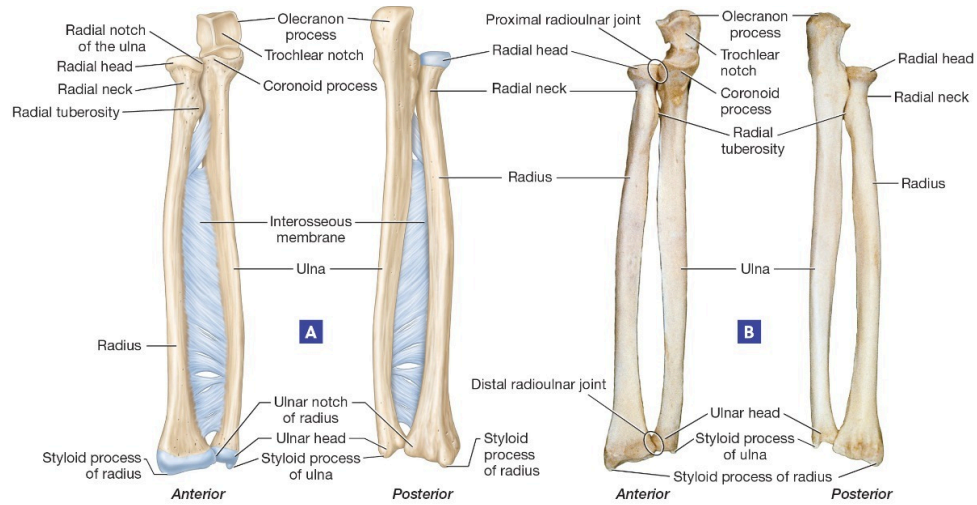


Figure 8.30 Right radius and ulna: (**A**) illustration, anterior and posterior views; (**B**) photograph, anterior and posterior views.

As its name implies, the trochlear notch fits around the trochlea of the humerus to form the elbow joint. The olecranon process is the actual "elbow bone," which you can feel on your posterior arm. The trochlear notch and the ulna's two processes form a "U" shape when the ulna is held on its side. This makes the ulna easy to differentiate from the radius (think "U" for "ulna").

The distal end of the ulna is known as the **ulnar head**. It articulates with one of the wrist (or *carpal*) bones that is known as the *lunate*. On its medial side, it has a small projection called the **styloid process** that is palpable through the skin.

The shape of the radius is opposite to that of the ulna: It is skinny proximally and wide distally. Proximally, it consists of a round, flattened **radial head** that articulates with the capitulum of the humerus to help form the elbow joint.

Distal to the radial head, we find the **radial neck** and a projection called the **radial tuberosity**. At its most distal end, the radius articulates with several carpal bones. It also features a lateral **styloid process**, just as we found on the ulna.

Notice in **Figure 8.31** that the radius and ulna also articulate with one another. On the ulna, just distal to the coronoid process, we find the **radial notch**, which is where the radius and ulna articulate at the **proximal radioulnar joint**. At the distal end of the radius is an indentation called the **ulnar notch** where the ulna fits into the radius. This joint is known as at the **distal radioulnar joint**. These joints are stabilized by the *interosseous membrane*, a band of dense regular collagenous connective tissue that joins the radius and ulna along their length.

The wrist is composed of eight short bones called **carpals** (KAR-pulz; *carp-* = "wrist"), which are arranged into two rows. The proximal row, from medial to lateral, consists of the *triquetrum*, *pisiform*, *lunate*, and *scaphoid* bones. The distal row, from medial to lateral, includes the *hamate*, *capitate*, *trapezoid*, and *trapezium* bones (**Figure 8.32**; remember that the thumb is lateral because we are in anatomical position). The carpals articulate proximally with the radius and the ulna, and distally with the five long bones in the hand called the **metacarpals**. The metacarpals articulate distally with the fingers, which are formed from 14 long bones called **phalanges** (fuh-LAN-jeez). The second through fifth digits have three phalanges each (the *proximal*, *intermediate* [or *middle*], and *distal phalanges*); the thumb has only two (a *proximal* and a *distal phalanx*).

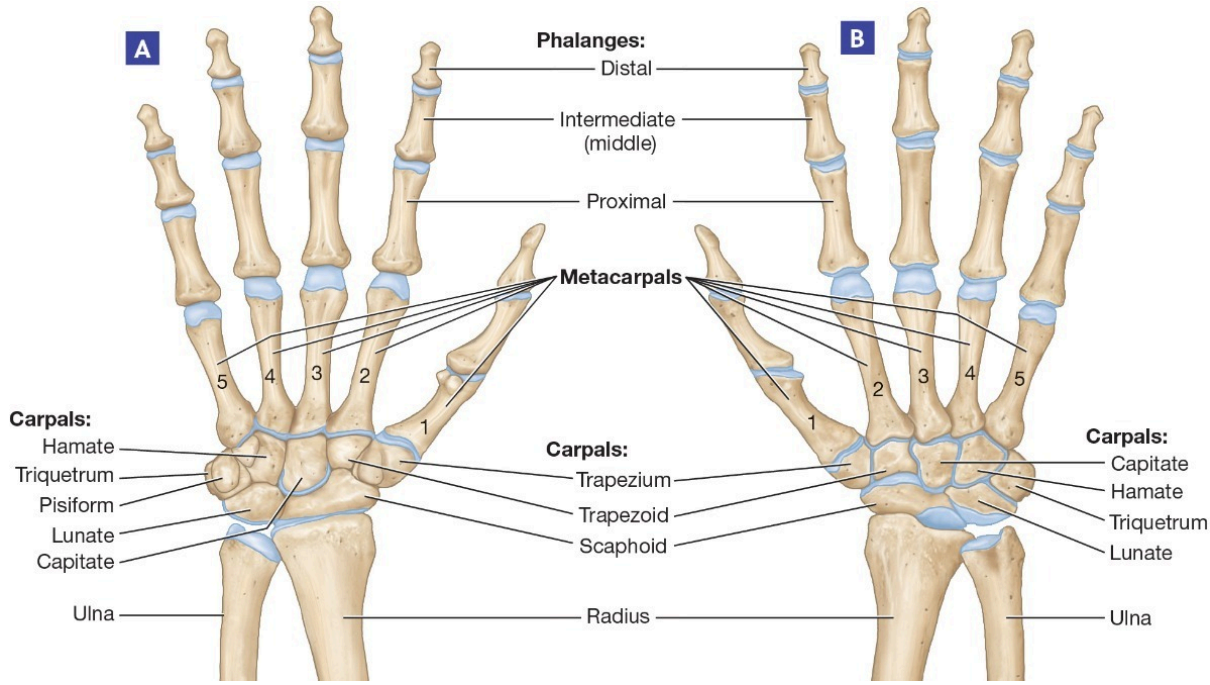


Figure 8.32 Right wrist and hand: (**A**) anterior view; (**B**) posterior view.

Pelvic Girdle

The **pelvis** (*pelvis* = "basin" or "bucket") connects the lower limbs to the trunk, supports the pelvic organs, and transmits the weight of the trunk to the legs. It is formed by the **pelvic girdle**, which itself is composed of two **coxal bones** (*cox-* = "hip;" **Figure 8.33**), the sacrum and the coccyx. The coxal bones meet the sacrum at the two sacroiliac joints. The opening into the pelvis is known as the **pelvic inlet**; the ridge around the inlet is the **pelvic brim**. The pelvic inlet and brim separate what is known as the **greater pelvis** (colored green in the figure), which sits superior to the pelvic inlet, from the **lesser pelvis** (colored yellow in the figure), which sits inferior to it. As you can see in **Figure 8.33B**, we find the **pelvic outlet** at the inferior edge of the lesser pelvis.

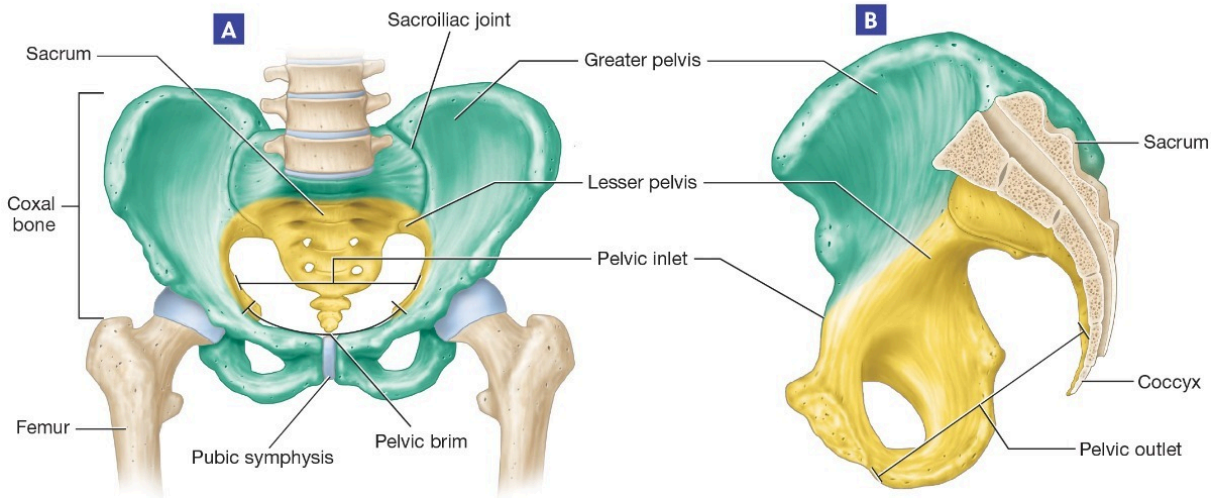


Figure 8.31 Pelvis: (**A**) anterior view; (**B**) midsagittal section.

Each coxal bone is also known as a **hemipelvis** (*hemi-* = "half"). Three fused bones—the ilium, the ischium, and the pubis—make up each coxal bone (**Figure 8.34**). You can see in the lateral view the place where all three bones come together to form a deep socket. This socket, called the **acetabulum** (ă-seh-TAB-yoo-lum), forms the hip joint with the femur. Notice also that where the ischium and pubis meet there is a large hole called the **obturator foramen** (AHB-too-ray-tur). In a living person, this hole is covered with a membrane and allows only small blood vessels and nerves to pass through.

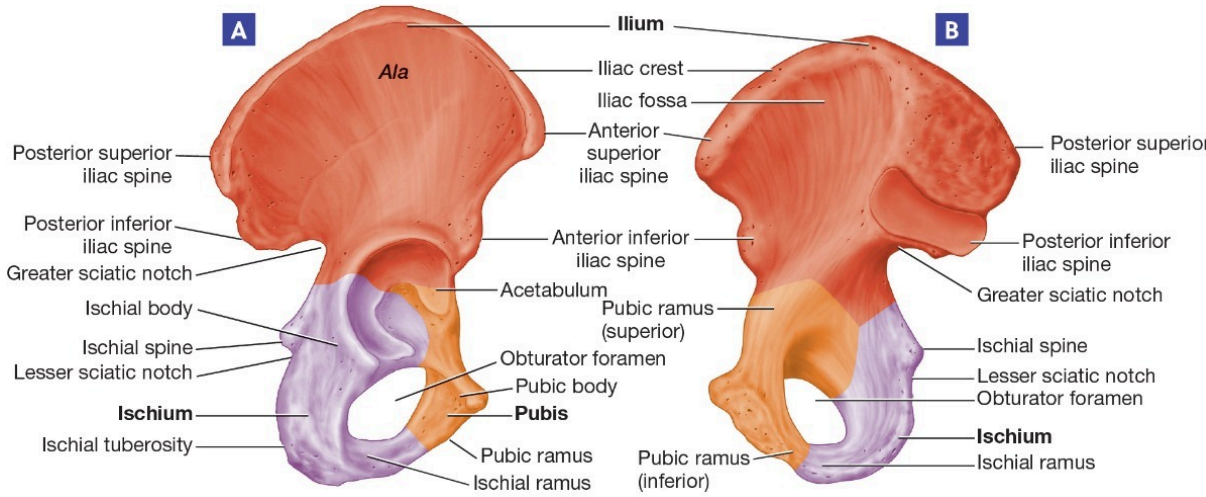


Figure 8.34 Hemipelvis: (**A**) right coxal bone, lateral view; (**B**) right coxal bone, medial view.

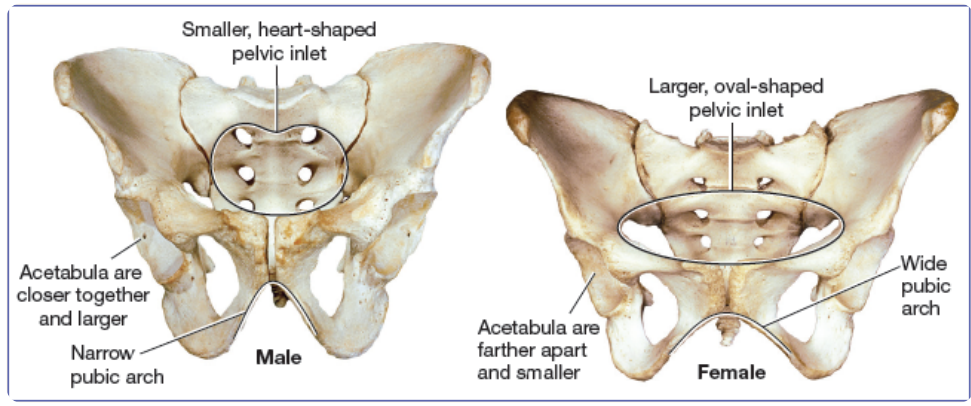
Let's look more closely at the three bones of the pelvic girdle and their features.

1. **Ilium**. The **ilium** (ILL-ee-um; *ilia-* = “flank”) is the largest of the three bones. Its main portion is called the **body**, and its superior “wing” is called the **ala** (AY-luh). The medial side of the ala is concave; this indentation is called the **iliac fossa**. The ridge of the ala, called the **iliac crest**, is where you rest your hands when your hands are on your hips. At the anterior end of the crest, we find a projection called the **anterior superior iliac spine,** and, inferior to it, the smaller **anterior inferior iliac spine**. A similar but smaller **posterior superior iliac spine** is located posteriorly, with a corresponding **posterior inferior iliac spine**. The posterior ilium also features a notch called the **greater sciatic notch** (sy-Ă-tik; *sciat-* = “hip joint”), which allows the greater sciatic nerve to pass from the pelvis to the thigh.
2. **Ischium**. The **ischium** (ISS-kee-um; *ischi-* = “hip joint”) makes up the posteroinferior pelvis. It contains three features on its posterior side: the superior **ischial spine**, the middle **lesser sciatic notch**, and the thick, inferior **ischial tuberosity**. The ischial tuberosities are the “butt bones”: the bones that bear your weight when you sit down. On the ischium’s anterior side, we find the **ischial ramus**, which articulates with the pubis.
3. **Pubis**. The anterior portion of the pelvis is formed by the **pubis** (PYOO-bis; *pub-* = “groin”) or the **pubic bone**. The pubis consists of a **body** and two extensions called the **superior** and **inferior rami**. The bodies of the two pubic bones meet at a fibrocartilage pad called the **pubic symphysis** (SIM-fih-sis; *sym-* = “together;” *-physis* = “growth;”see **Figure 8.33**). The angle that these bones form when they meet, called the **pubic arch**, can help to determine the sex of a skeleton (see the “Hints and Tips” box).

Hints and Tips : How to Determine Sex Based upon Pelvic Features

A skeleton's sex can be determined based upon the differences between the male and female pelvises. In general, the male and female pelvis can be distinguished by the features listed below and shown in the photos here:

Feature	Male Pelvis	Female Pelvis
Pelvic inlet shape	Narrower and heart-shaped	Wider and oval-shaped
Pubic arch	Narrow angle	Wide angle
Acetabulae	Closer together	Farther apart
Ischial tuberosities	Inverted	Everted
Coccyx	Curved anteriorly, less movable	Straighter, more movable



Pelvic bones, male and female

Lower Limb

The lower limb consists of the thigh, the **patella** or kneecap (puh-TEL-uh; "little plate"), the leg, the ankle, and the foot. The thigh contains only a single bone—the large, heavy **femur** (**Figure 8.35**). Proximally, the femur articulates with the acetabulum at its rounded **head**. In the center of the femoral head, we find a pit called the **fovea capitis** (FOH-vee-uh CAP-ih-tiss). This is the point of attachment for a ligament that holds the femur in the acetabulum. Just distal to the femoral head is the **neck** of the femur, the weakest part of the femur and

the most common location for it to fracture (note that when the femoral neck fractures, it is usually called a “broken hip,” even though the hip itself doesn’t fracture). Where the femoral neck meets the femoral shaft, we find two large prominences: the anterolateral **greater trochanter** (TROH-kan-tur) and the posteromedial **lesser trochanter**. Between the two trochanters on the anterior side is a ridge called the **intertrochanteric line** (in-ter-troh-kan-TEHR-ik), and on the posterior side is another ridge in the same location known as the **intertrochanteric crest**.

The femoral diaphysis is mostly smooth, save for a few features that mark the attachment points for specific muscles on the posterior side:

- The proximal **gluteal tuberosity** (GLOO-tee-uhl; *glut-* = “buttocks”) is a rough projection that serves as the attachment spot for one of the gluteal muscles (the gluteus maximus muscle).
- The **linea aspera** (LIN-ee-ah ASS-per-ah) is a prominent line that runs along the femoral diaphysis. It serves as an attachment point for many muscles that move the femur.
- Near the femoral epiphysis, the linea aspera splits into the **medial** and **lateral supracondylar lines** (soo-prah-KAHN-duh-lar), where hamstring muscles attach to the femur.

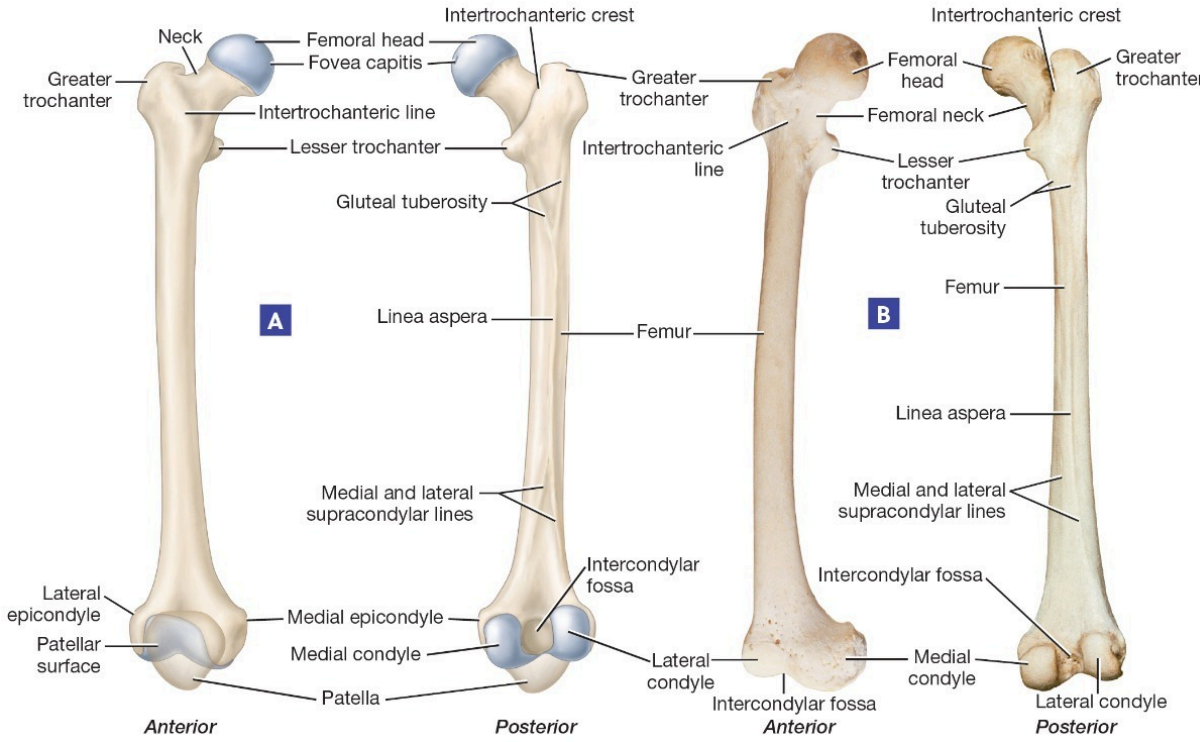


Figure 8.35 Right femur: (**A**) illustration, anterior and posterior views; (**B**) photograph, anterior and posterior views.

Distally, the femur expands into two small projections, the **medial** and **lateral epicondyles** (ep-ih-KAHN-dyl'z), distal to which we find the large **medial** and **lateral condyles**, which form the knee joint with the largest bone of the leg—the tibia. On the posterior side of the femur is a depression between the two condyles, the **intercondylar fossa**.

The **tibia** (TIB-ee-uh) is the medial and larger of the two leg bones (**Figure 8.36**). It is flattened proximally at its articular surface, the **tibial plateau**, where its **medial** and **lateral condyles** fit together with those of the femur. Between the medial and lateral condyles is a ridge called the *intercondylar eminence*. Just distal to the condyles on the tibia's anterior surface is a rough projection called the **tibial tuberosity**, which is where the patellar ligament inserts. Along the anterior tibial diaphysis is a sharpened ridge known as the **anterior crest**, which is commonly known as the "shin." At the tibia's distal epiphysis, it articulates with a tarsal bone called the *talus*, with which it forms the ankle joint. At its terminal end is a projection called the **medial malleolus** (mal-ee-OH-luhs), which you may think of as your "medial ankle bone."

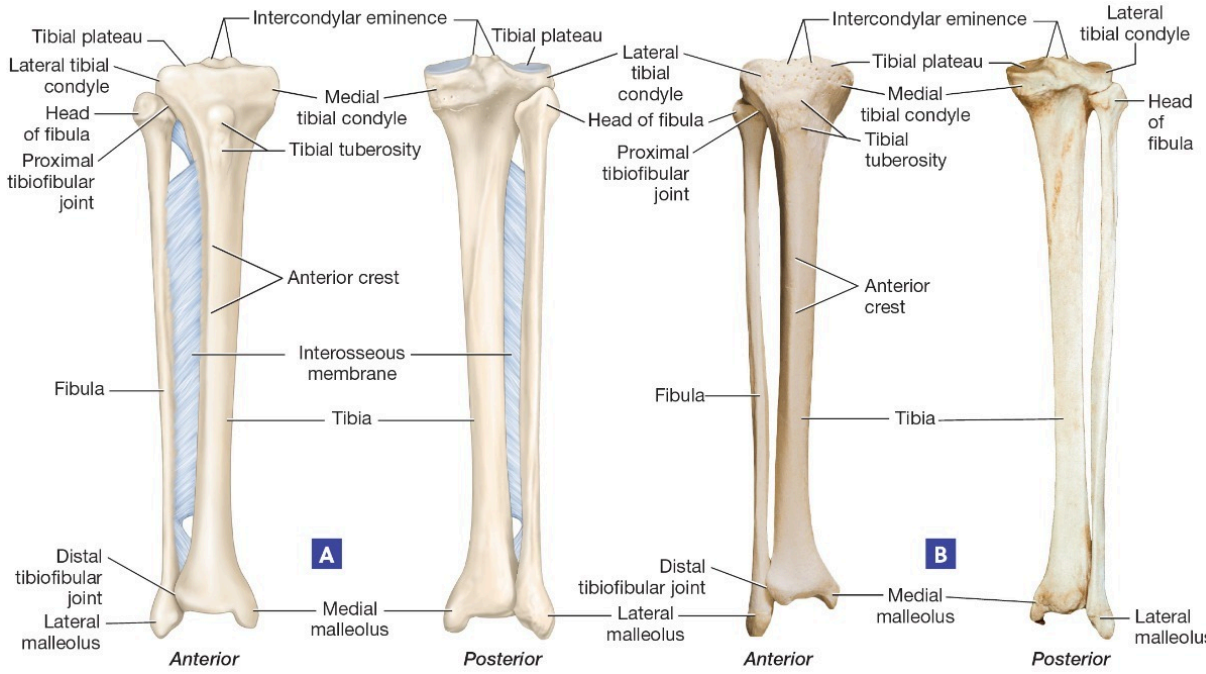


Figure 8.36 Right tibia and fibula: (**A**) illustration, anterior and posterior views; (**B**) photograph, anterior and posterior views. posterior views.

The other leg bone is the thin, lateral **fibula** (FIB-yoo-luh). The proximal fibular end is called the **head**, and its distal end is the **lateral malleolus** (which you may think of as your "lateral ankle bone"). The head has no articulation with the femur, but the lateral malleolus articulates with the talus to help form the ankle joint.

Notice in **Figure 8.36** that the tibia and fibula articulate with one another just as the radius and ulna do. At the proximal end, we find the **proximal tibiofibular joint**, and, at the distal end, we find the **distal tibiofibular joint**. The two bones are also held together by a stabilizing interosseous membrane.

The ankle is composed of seven short bones called **tarsals** (TAHR-sulz; *tars-* = "ankle;" Figure 8.35). The largest tarsal bone is the **calcaneus** (kal-KAYN-ee-uhs; *calcan-* = "heel"), or the heel bone. Resting on top of the calcaneus is the **talus** (TAY-luhs), which forms the ankle joint with the tibia and fibula. Distally, the talus articulates with a ship-shaped bone called the *navicular*. The remaining tarsals are, from medial to lateral, the *medial cuneiform*, *intermediate cuneiform*, *lateral cuneiform*, and *cuboid* bones. The tarsals articulate distally with the five long bones in the foot called **metatarsals** (met-

uh-TAHR-sulz). Notice in **Figure 8.37** that there are two small *sesamoid bones* at the head of the first metatarsal—these bones are found in a tendon located here. Like the bones of the fingers, the bones of the toes also consist of 14 **phalanges**. The second through fifth digits have three phalanges each (the *proximal*, *intermediate* [or *middle*], and *distal phalanges*); and the big toe or *hallux* has only two (a *proximal* and a *distal phalanx*).

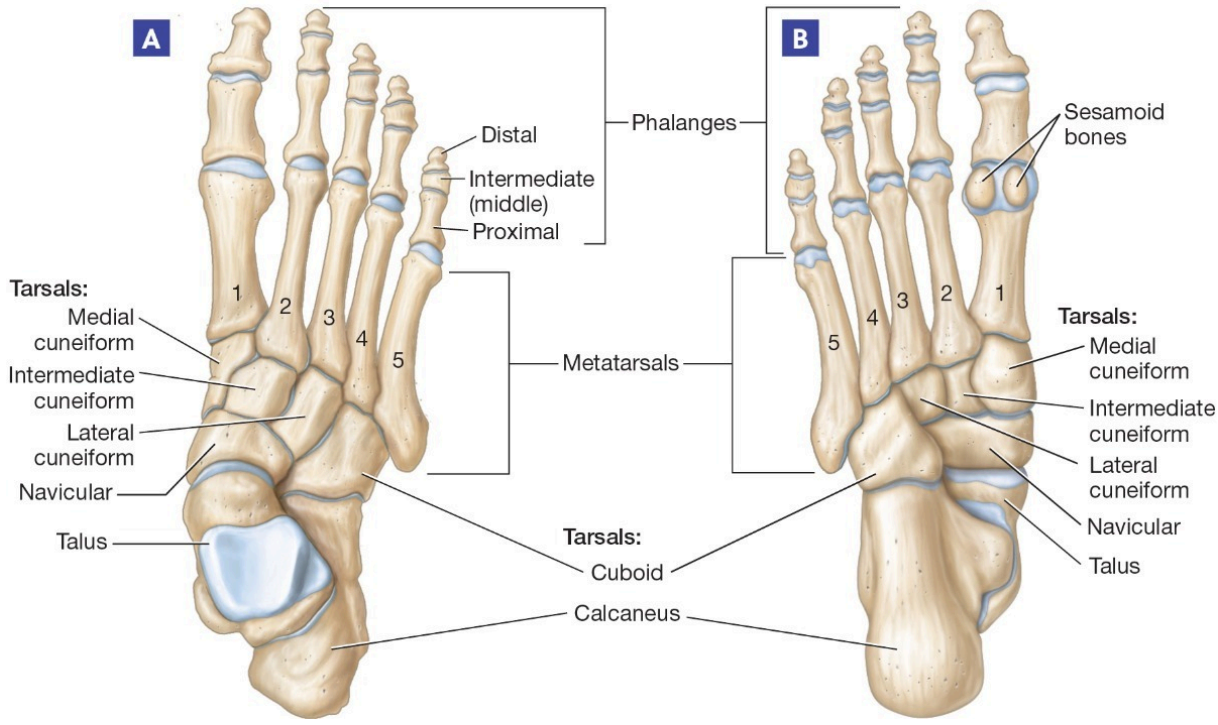


Figure 8.37 Right foot and ankle: (**A**) superior view; (**B**) inferior view.

Procedure 2: Cooperative Learning for the Axial and Appendicular Skeletons

We will follow essentially the same procedure here as we did in Exercise 8-1, but with different groups of bones and bone markings. Note that this cooperative learning exercise includes the structures from Exercises 8-2 and 8-3.

1. Assemble into teams with a minimum of four students; five is optimum.

2. Distribute a bone or set of bones to each member of the team, and assign each student one of the following five groups of bones and bone markings (the

specific structures of each bone are listed below):

- *Group A*: Cervical, thoracic, and lumbar vertebrae and vertebral markings; ribs and rib markings; sternum structures; the hyoid bone.
- *Group B*: Scapula structures; clavicle structures; humerus structures.
- *Group C*: Radius and ulna markings; carpals, metacarpals, and phalanges.
- *Group D*: Ilium, ischium, and pubis structures; the difference between the male and female pelvises.
- *Group E*: Femur, tibia, and fibula structures; tarsals, metatarsals, and phalanges.

3. Spend approximately 3 minutes learning the assigned structures on your own.

4. Spend about 1 to 2 minutes teaching your assigned structures to your team. Then have each team member spend approximately 1 to 2 minutes teaching the team his or her assigned structures.

5. Rotate the assigned structures clockwise so each member has a new set of structures to learn and teach (the student that was assigned Group A will take Group B, and so on).

6. Repeat Steps 3 and 4, and then rotate assigned structures again. Continue this process (it begins to speed up significantly at this point) until each team member has taught each group of structures once.

The following is a list of bone and bone markings of the axial and appendicular skeletons covered in this exercise.

Remaining Structures of the Axial Skeleton

Group A Structures

1. Vertebral column

- **a.** Intervertebral foramina
- **b.** Cervical curvature
- **c.** Lumbar curvature
- **d.** Thoracic curvature
- **e.** Sacral curvature

2. Vertebrae

- **a.** Spinous process
- **b.** Transverse processes
- **c.** Vertebral foramen
- **d.** Vertebral body
- **e.** Intervertebral disc
- **f.** Pedicle
- **g.** Lamina

3. Cervical vertebrae

- **a.** Transverse foramina
- **b.** Atlas
- **c.** Axis
- **d.** Dens

4. Thoracic vertebrae

- **a.** Costal (articular) facets

5. Lumbar vertebrae

6. Sacrum

- **a.** Sacral promontory
- **b.** Sacral canal
- **c.** Anterior sacral foramina
- **d.** Posterior sacral foramina

7. Coccyx

8. Sternum

- **a.** Manubrium
- **b.** Body
- **c.** Xiphoid process

9. Ribs

- **a.** Intercostal spaces
- **b.** Costal cartilage
- **c.** True (vertebrosternal) ribs
- **d.** False ribs
  - (1) Vertebrochondral ribs
  - (2) Floating ribs (vertebral ribs)
- **e.** Costal margin
- **f.** Head of rib
- **g.** Neck of rib
- **h.** Shaft of rib

10. Hyoid bone

Pectoral Girdle and Upper Limb

Group B Structures

1. Scapula

- **a.** Superior, lateral, and medial borders
- **b.** Superior, lateral, and inferior angles
- **c.** Subscapular fossa
- **d.** Glenoid cavity
- **e.** Coracoid process
- **f.** Acromion
- **g.** Spine

h. Supraspinous fossa